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Bachelor Thesis

“Non-linear Dynamical Systems with Chaotic Behaviour: an application on rumour spreading on social media”

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SUMMARY

This thesis studies non-linear dynamical systems exhibiting chaotic behaviour, with the aim of developing a theoretical framework and demonstrating its application to a problem of social relevance. The first part introduces the basic concepts of dynamical systems theory. Next, it focuses on the main concepts of chaos theory, such as the study of the Lorenz system, sensitive dependence on initial conditions, Lyapunov exponents, the logistic map, Feigenbaum universality, fractal geometry, and strange attractors. The second part proposes a seasonally forced SEIR model for the spread of rumours on social media platforms, in which the effect of recommendation algorithms is considered. The model is analysed computationally using Python, through bifurcation diagrams, maximal Lyapunov exponent calculations, and phase portrait analysis across different parameter configurations. This allows us to characterise the conditions under which the system transitions from stable to chaotic dynamics, providing insights on how user behaviour and algorithmic forcing govern the spread of information online.

Keywords: chaos theory, non-linear dynamical systems, bifurcation theory, Lyapunov exponents, Feigenbaum universality, strange attractor, SEIR model, rumour spreading dynamics.

DEDICATION

First of all, I would like to express my gratitude to my parents, and especially to my stepfather, who has shared with me his curiosity and admiration for mathematics and science for as long as I can remember.

To all the mathematics teachers who taught me during my school years. Without such dedicated teachers, I would never have developed the level of interest in mathematics that I have today. I also want to express my admiration for my university professors, and in particular for Froilán, who told me in my first year that I was going to become a great mathematician.

I would also like to give special thanks to Pablo, my thesis supervisor. He has supported me from the very first moment I proposed the idea, through the days when I was more and less motivated, and has given me the confidence not to doubt myself.

To my stepsisters, for being a wonderful example to follow and a fundamental source of support.

But above all, I want to end by thanking my university friends, who have made these years the best of my life. For always being there for me, for encouraging me when I needed it, for helping me when I was lost, and for bringing a smile to my face every single day.

I am deeply grateful for all the lessons and moments that I will carry with me from this chapter of my life.

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1. INTRODUCTION

This thesis carefully explores the topic of *non-linear dynamical systems*, specifically those exhibiting *chaotic behaviour*. Dynamical Systems Theory is the branch of mathematics responsible for analysing how a process changes over time; in other words, it explores the evolution of processes. These processes happen everywhere in nature. For this reason, not only mathematicians, but also physicists, meteorologists, economists, biologists, and scientists from many other fields find relevance in this topic. Expanding the acquired concepts in the course on *Ordinary Differential Equations*, this project focuses on the study of the main theoretical concepts of Chaos Theory, together with examples and applications by simulating chaotic behaviour. In particular, the theoretical framework developed in the first two chapters is put into practice in Chapter 4, where an original forced SEIR model for the spread of rumours on social media platforms is proposed and analysed computationally.

1.1. Motivation

The butterfly effect is one of those mathematical ideas that has become part of everyday language. The fact that a small, almost imperceptible shift in the present can drastically alter future outcomes reflects a human reality. A chance encounter, a small decision, any of these can lead to a sequence of outcomes whose final consequences are impossible to predict. This intuition, which for centuries was little more than a philosophical observation, turns out to be a rigorous mathematical phenomenon with a beautiful theory behind it.

This thesis grew out of a desire to understand that theory properly. As a mathematics student, the study of ordinary differential equations raises natural questions about what happens when systems become non-linear. Chaos theory offers an answer to this question, and the mathematical tools it has developed are both powerful and elegant. The desire to explore these tools in depth, and to see them applied to a problem of social relevance, is what motivated the work presented here.

A particular example of social relevance is the spread of information on social media. Today, a rumour can reach millions of users within hours, driven by recommendation algorithms that amplify content in ways that are difficult to predict or control. Understanding when and why this process becomes chaotic is both a mathematical question and a matter of real social concern, and it is the problem that Chapter 4 of this thesis addresses.

1.2. Objectives

The main objectives of this thesis are the following.

The first objective is to introduce the theoretical foundations of non-linear dynamical systems and chaos theory, including the Lorenz system, the logistic map, Lyapunov exponents, and fractal geometry. These topics are used as conceptual tools for understanding and analysing chaotic behaviour in dynamical systems.

The second objective is to propose a model for the spread of rumours on social media platforms, based on a seasonally forced SEIR system, and analyse its chaotic properties. This model incorporates the periodic amplification behaviour of recommendation algorithms, and is designed to capture the complex dynamics that arise from the interaction between user behaviour and algorithmic forcing.

The third objective is to simulate the chaotic behaviour of the model computationally using Python, analysing bifurcation diagrams, maximal Lyapunov exponents, and phase portraits across different parameter configurations, in order to characterise the conditions under which the system transitions from stable to chaotic dynamics.

1.3. Document structure

This thesis is organised as follows. Chapter 2 introduces the main concepts of dynamical systems theory, including iterated maps, differential equations, stability analysis, bifurcations, and limit cycles. Chapter 3 reviews the core topics of chaos theory: the Lorenz system, sensitive dependence on initial conditions, the logistic map, Feigenbaum universality, fractal geometry, and strange attractors. Chapter 4 presents the proposed forced SEIR model for rumour spread on social media, together with a detailed computational analysis of its bifurcation structure, Lyapunov exponents, and phase portraits. Chapter 5 addresses the regulatory framework and socioeconomic impact of the work. Chapter 6 presents the project planning and budget. Finally, Chapter 7 concludes the thesis with a summary of the main findings and a discussion of future research directions.

All the figures in this thesis (except the schematic ones) have been generated using Python 3, primarily with the libraries `NumPy`, `SciPy`, and `Matplotlib`. The corresponding scripts are available in a Github repository, which can be found in Appendix A.

1.4. Brief history

For centuries, the understanding of dynamical systems was shaped by the *deterministic Laplacian paradigm*, derived from the laws of classical Newtonian physics. This paradigm is based on the idea that if one could know the exact position and momentum of every particle in the universe at a given instant, the laws of mechanics would allow to compute its whole future with accuracy. Thanks to Newton's work, differential equations became the principal mathematical tool to describe dynamical processes continuous in time. Different techniques were developed for solving these equations, although most of them were useful only for linear systems.

However, *Chaotic Systems* did not fit within this framework. It is often said that classical science ends where chaos begins. Chaos theory rejects the idea that simple systems necessarily yield simple solutions: they are now understood as capable of producing complex predictability issues. This implies that deterministic systems can be unpredictable in the long run.

Although there are many ways to describe chaos in dynamical systems, they are all grounded in the phenomenon known as *sensitive dependence on initial conditions*. This phenomenon refers to the observation that imperceptible variations in the input lead to vastly divergent outcomes. Due to this nature, physicists initially considered the topic too abstract to deal with, while mathematicians viewed the emerging computational evidence as lacking formal rigor.

Consequently, chaotic behaviour was often ignored by the scientific community because it could not be analysed with the linear tools of the time.

The first major crack in the Laplacian paradigm appeared in the late 19th century with **Henri Poincaré**. A prize was announced for the first scientist to solve the *n-body problem*, which was unsolvable for $n \geq 3$. This consisted of a system of non-linear differential equations whose solutions described the motion of n point masses moving in space subject only to their mutual gravitational attraction. While studying the *Three-Body Problem*, Poincaré used geometric and topological techniques, questioning the assumption that systems could only be stable or unstable. He realised that even simple deterministic systems could behave in non-integrable, wildly erratic ways, making long-term prediction impossible. He discovered what we now call *chaos*.

Despite Poincaré's early warnings, the field only truly flourished when computers became available to perform the necessary calculations. During the 1960s, non-linear oscillators were playing a vital role in the study of dynamics. **Stephen Smale**'s studies on this topic led to important propositions on non-linear chaotic oscillators. By representing the complexity of Van der Pol's oscillator using topological transformations, he developed the *Smale horseshoe*, which structurally characterised chaotic behaviour.

The real revolution started when high-speed computers allowed experimentation on dynamical systems. In 1963, meteorologist **Edward Lorenz** rediscovered chaotic behaviour while modelling atmospheric convection. He observed that small rounding errors in his computer simulations led to completely different weather patterns, a phenomenon that later became popularly known as the *butterfly effect*. Lorenz visualised these solutions in phase space. He found that the system did not settle into a steady state or a periodic loop. Instead, the trajectories traced a complex, non-repeating pattern that stayed confined within a finite region. This geometric structure, was the first example of a **strange attractor**.

Later, in 1975, **Tien-Yien Li** and **James Yorke** proved that every unidimensional system where period 3 appeared (meaning the existence of a cycle in which the system returns to its initial state after exactly three iterations) implied chaotic behaviour in their seminal paper *Period Three Implies Chaos* (see [13]). These achievements had a great impact on different areas of science. In ecology, **Robert May** studied the logistic equation, a simple model designed to predict how a population evolves over time given a growth rate r. He observed that, as this growth rate increases, the population shifts from stable behaviour (which means converging to a fixed value) to oscillatory behaviour (which means alternating between two or more distinct values) and eventually to chaos. This model will be studied in detail in Section 3.3.

The diverse studies on chaotic behaviour raised the question of how this could be considered a universal law for different systems. Physicist **Mitchell Feigenbaum** answered this with the theory of universality, proposing that different systems turn chaotic following a universal pattern. The last great discovery occurred thanks to **Benoit Mandelbrot**, whose studies completely changed traditional geometric thinking by introducing fractal dimensions, focusing on scale and roughness rather than smooth shapes. This concept became fundamental for dynamics, as *strange attractors* turn out to have fractional dimensions. In Section 3.4 we will study this concept in depth, analysing how fractal geometry arises in chaotic systems.

Chaos theory has opened infinite possibilities for the scientific community: a new way of thinking about the world around us, a new way in which scientists analyse nature, and a completely new paradigm for mathematicians.

To know more about the historical context of Chaos Theory, see [8].

2. BASIC CONCEPTS OF DYNAMICAL SYSTEMS

In this chapter, the main concepts about *dynamical systems* will be defined, with definitions, examples and exercises from [21].

A dynamical system can be represented either by differential equations, if the system is time-continuous, or by iterated maps, if the system is time-discrete.

Iterated maps

Many systems are described by difference equations, where time jumps in discrete steps ($n = 0, 1, 2, \dots$). These are called *iterated maps*:

$$x_{n+1} = f(x_n). \tag{2.1}$$

Here, the state x_n determines the next state x_{n+1}.

Definition. A *fixed point* for a map f is a point x^* such that $f(x^*) = x^*$.

Example. The map $f(x) = x^3$ has exactly three fixed points: 0, 1, and -1 .

Definition. A point x is a *periodic point* of period n if $f^n(x) = x$, where f^n denotes the n-th composition of f with itself. The least positive integer n for which this equation holds is called the *prime period* of x .

Example. For the map $f(x) = -x$, the origin is a fixed point, while all other real numbers have a prime period of 2 .

Stability of fixed points

In order to determine the stability of a fixed point, we consider a small perturbation η_n away from the fixed point:

$$\eta_n = x_n - x^*. \tag{2.2}$$

We want to see whether this perturbation grows or decays as the map is iterated. To find the evolution of η_n, we apply the map and use the Taylor series expansion of $f(x)$ around x^*:

$$x_{n+1} = f(x_n) \implies x^* + \eta_{n+1} = f(x^* + \eta_n). \tag{2.3}$$

Expanding the right-hand side, we get:

$$x^* + \eta_{n+1} = f(x^*) + f'(x^*)\eta_n + O(\eta_n^2). \tag{2.4}$$

Since x^* is a fixed point, we know that $f(x^*) = x^*$. Substituting this into the equation and assuming that the perturbation η_n is very small, we can get rid of higher-order terms $O(\eta_n^2)$. This gives us the equation:

$$\eta_{n+1} \approx f'(x^*)\eta_n. \tag{2.5}$$

The derivative evaluated at the fixed point, $\lambda = f'(x^*)$, is often called the *multiplier* or *eigenvalue* of the fixed point. Therefore the local stability of x^*, is determined by the absolute value of λ:

- If $|\lambda| < 1$, the perturbation $\eta_n \to 0$ as $n \to \infty$. The fixed point x^* is **stable** (or an *attracting* fixed point).
- If $|\lambda| > 1$, the perturbation grows exponentially with n. The fixed point x^* is **unstable** (or a *repelling* fixed point).
- If $|\lambda| = 1$, the linear approximation $\eta_{n+1} \approx \lambda\eta_n$ is inconclusive. In this case, the non-linear terms $O(\eta_n^2)$ determine the stability of the fixed point.

Remark. The sign of the multiplier λ also gives us geometric information about how the trajectories behave near x^*. If $0 < \lambda < 1$, the sequence approaches x^* monotonically. If $-1 < \lambda < 0$, the perturbation alternates in sign at each step, meaning the state x_n spirals or oscillates towards x^*.

Example. Consider the map given by $f(x) = x^2$. To find its fixed points, we set $f(x^*) = x^*$, which yields the equation:

$$(x^*)^2 = x^* \implies x^*(x^* - 1) = 0. \tag{2.6}$$

Thus, there are two fixed points: $x^* = 0$ and $x^* = 1$.

To determine their stability, we compute the derivative of the map, $f'(x) = 2x$, and evaluate the multiplier λ at each fixed point:

- For $x^* = 0$, the multiplier is $\lambda = f'(0) = 0$. Since $|\lambda| < 1$, the origin is a stable fixed point.
- For $x^* = 1$, the multiplier is $\lambda = f'(1) = 2$. Since $|\lambda| > 1$, this is an unstable fixed point.

Graphical analysis: cobweb diagrams

There exist useful tools for analysing iterative maps graphically. The *cobweb diagram* is used to trace the orbit of an iterated map $x_{n+1} = f(x_n)$ without needing to compute the numerical values of each step explicitly. This allows us to observe the behaviour of the map in a more visual way.

The construction of a cobweb plot follows a simple algorithm:

1. Draw the curve representing the function $y = f(x)$ that corresponds with $f(x_n)$ and the diagonal line $y = x$ on the same plot. Note that the intersections of these two graphs correspond exactly to the fixed points of the map, since at these points $f(x^*) = x^*$.
2. Choose an initial condition x_0 on the x-axis.
3. Draw a vertical line from $(x_0, 0)$ to the curve $y = f(x)$. This intersection point is $(x_0, f(x_0))$, which gives us the value of the first iteration, x_1.
4. From this point on the curve, draw a horizontal line to the diagonal $y = x$. This intersection point is (x_1, x_1). This translates our new output value x_1 back into an input value for the next iteration.
5. From the diagonal, draw a vertical line back to the curve to find the next state $(x_1, f(x_1))$, which is (x_1, x_2).
6. Repeat the horizontal and vertical movements indefinitely to trace the orbit.

This iterative process creates a path of alternating horizontal and vertical line segments that allow us to check whether the system converges to a fixed point or diverges going away from it.

Example. We can check graphically the behaviour of the fixed points over the map previously analysed, $f(x) = x^2$. The diagram is constructed by iterating the map numerically from a chosen initial condition x_0. For the left panel, the initial condition is set to $x_0 = 0.8 \in (0, 1)$, and the map is iterated for $n = 10$ steps, which is sufficient to observe clear convergence towards the stable fixed point $x^* = 0$. For the right panel, the initial condition is $x_0 = 1.05$, slightly above the unstable fixed point $x^* = 1$, and $n = 5$ iterations suffice to show the divergence of the orbit before it escapes the plotted domain. In both cases, the curve $y = f(x) = x^2$ and the diagonal $y = x$ are evaluated on a uniform grid of 500 points over the respective domains.

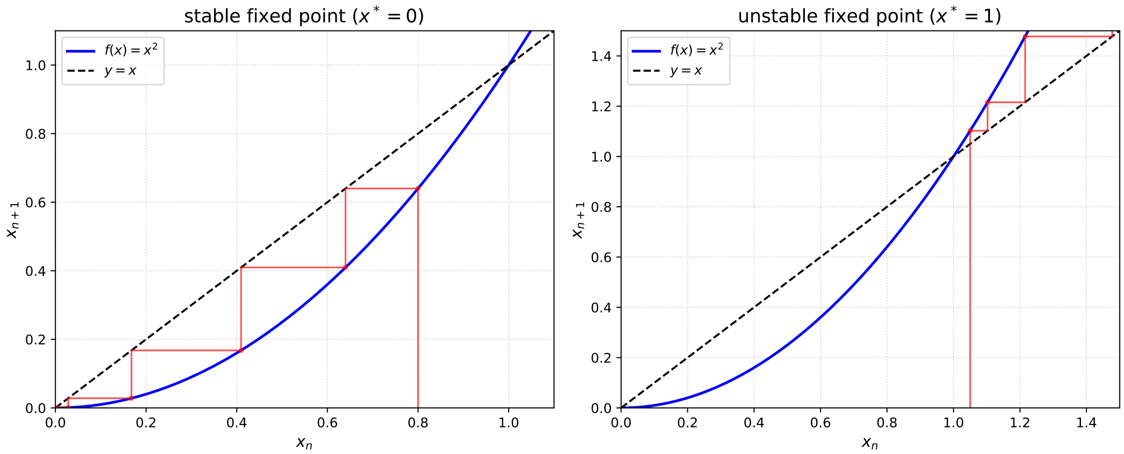

Fig. 2.1. Cobweb diagram for the map $f(x) = x^2$.

As we can observe in 2.1, for values close to $x^* = 0$, (specifically those in the interval $(-1, 1)$), the system converges to this fixed point. On the other hand, for values strictly bigger than 1, (those on $(1, \infty)$)), the system diverges from $x^* = 1$.

One-dimensional maps can exhibit chaotic behaviour. This makes them powerful tools for studying chaos in a simplified setting. A famous example is the *logistic map*, which we will study in Chapter 3.

Differential equations

These equations describe dynamical systems continuous in time.

Definition. A *differential equation* is an equation of functions where the unknown is a function.

A very general framework for ordinary differential equations is provided by the system:

$$\begin{aligned} \dot{x}_1 &= f_1(x_1, \ldots, x_n) \\ &\vdots \\ \dot{x}_n &= f_n(x_1, \ldots, x_n). \end{aligned}$$

Remark. The dots over the variable represent differentiation with respect to time.

Definition. An *ordinary differential equation* (ODE) is an equation involving an unknown function of one independent variable and its derivative.

Example. A classic physical example of an ordinary differential equation is Newton's second law of motion:

$$F(t) = m\ddot{x}(t).$$

Definition. A *partial differential equation* (PDE) is an equation involving an unknown function of multiple independent variables and its partial derivatives.

Example. A fundamental example of a partial differential equation is the heat equation:

$$\frac{\partial u}{\partial t} = \alpha\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right).$$

Definition. An ordinary differential equation is said to be *linear* if it has the form:

$$a_n(t)u^{(n)}(t) + \cdots + a_1(t)u'(t) + a_0u(t) = f(t).$$

where the coefficients $a_i(t)$ and the term $f(t)$ depend only on the independent variable t.

Example. An example of a linear ODE is the equation of motion with resistive force:

$$v' = P - kv.$$

Definition. An equation that is not linear is called *non-linear*.

Example. The equation representing the conservation of energy is non-linear:

$$\frac{1}{2}m(\dot{x})^2 + V(x) = E.$$

Here, the non-linearity arises from the squared derivative term $(\dot{x})^2$.

2.1. One-dimensional systems

To understand the long-term behaviour of time continuous dynamical systems without solving the differential equation analytically, we use geometric analysis. We will introduce the main concepts by analysing the following non-linear system:

$$\dot{x} = f(x) = \sin(x). \tag{2.7}$$

Definition. The *stationary points* of a system are the values x^* such that $\dot{x} = 0$. At these points, there is no flow, and the state of the system remains constant for all time.

To find the stationary points of our example, we set the vector field $f(x)$ to zero:

$$\sin(x) = 0,$$

The solutions are the integer multiples of π. Therefore, we have an infinite number of stationary points given by:

$$x^* = k\pi, \quad \text{where } k \in \mathbb{Z} \quad .$$

Remark. Stationary points are also called *fixed points*.

Definition. A stationary point x^* is *stable* (or an attractor) if all nearby trajectories flow towards it. It is *unstable* (or a repellor) if nearby trajectories flow away from it.

Mathematically, we can determine linear stability by looking at the derivative of the vector field, $f'(x)$. If we introduce a small perturbation $\eta = x - x^*$ near the stationary point, its evolution is governed by $\dot{\eta} \approx f'(x^*)\eta$. The sign of $f'(x^*)$ therefore controls whether this perturbation grows or decays over time:

- If $f'(x^*) < 0$, the flow points towards the stationary point and the perturbation decays, so the point is **stable**.
- If $f'(x^*) > 0$, the flow points away from the stationary point and the perturbation grows, so the point is **unstable**.

Applying this to our example, where $f'(x) = \cos(x)$:

1. For even k :
$$f'(2n\pi) = \cos(2n\pi) = 1 > 0.$$
These points are unstable. The flow moves away from them.
2. For odd k:
$$f'((2n+1)\pi) = \cos((2n+1)\pi) = -1 < 0.$$
These points are stable. The flow converges towards them.

Definition. A *trajectory* (or orbit) is the specific solution curve $x(t)$ that describes the evolution of the state of the system over time, starting from a given initial condition $x(0) = x_0$. Geometrically, it represents the motion of a point flowing along the vector field.

Definition. The *phase portrait* is the geometric representation of all possible trajectories of the system. In one dimension, it consists of the real line with arrows indicating the direction of flow.

For $\dot{x} = \sin(x)$, the flow direction is determined by the sign of the sine function. The diagram is constructed over the domain $x \in [-\frac{5\pi}{2}, \frac{5\pi}{2}]$, discretised with 500 equally spaced points, which is sufficient to display the five fixed points $x^* \in \{-2\pi, -\pi, 0, \pi, 2\pi\}$ within the plotted range. Unstable fixed points (even multiples of π) are marked with open red circles, and stable fixed points (odd multiples of π) with filled green circles, consistent with the stability analysis above. The flow arrows are placed at representative interior points of each interval between consecutive fixed points, with their direction determined by the sign of $\sin(x)$ on that interval. We can see the graphical representation in Figure 2.2:

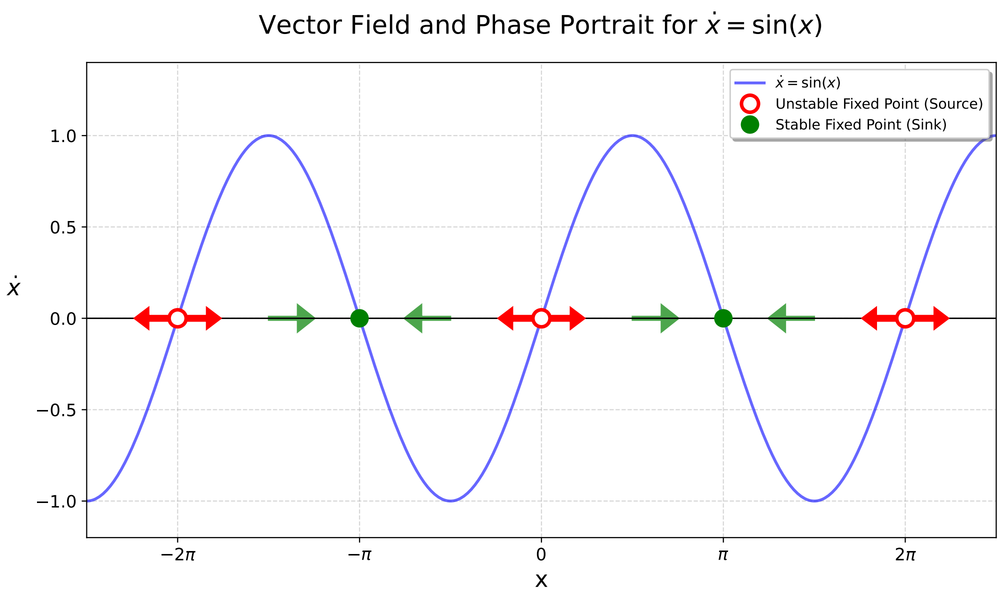

Fig. 2.2. Vector field and phase portrait for the system $\dot{x} = \sin(x)$.

Dynamical systems are often dependent on parameters. As a control parameter is varied, the qualitative structure of the flow can change. We call these changes *bifurcations*.

A classic example (see [21] section 3.4 for further information) is the *supercritical pitchfork bifurcation*. The normal form is given by:

$$\dot{x} = rx - x^3. \tag{2.8}$$

Here, r is the control parameter. To find the fixed points, we solve $\dot{x} = 0$:

$$x(r - x^2) = 0,$$

The stability analysis now depends on the parameter r:

1. Case $r < 0$: The term $(r - x^2)$ is always negative, so the only real fixed point is the origin $x^* = 0$. At the origin, $f'(0) = r < 0$. Thus, the origin is a **stable** fixed point.

2. Case $r = 0$: The equation becomes $\dot{x} = -x^3$. The only fixed point is still $x^* = 0$. However, $f'(0) = 0$. Linear stability analysis is inconclusive. By analysing the non-linear flow directly, we see that for $x > 0$, $\dot{x} < 0$ (flow to the left), and for $x < 0$, $\dot{x} > 0$ (flow to the right). Therefore, the origin remains **stable**.

3. Case $r > 0$: The origin $x^* = 0$ is still a fixed point, but now $f'(0) = r > 0$, making it **unstable**. Simultaneously, two new fixed points emerge at $x^* = \pm\sqrt{r}$. Evaluating stability at these points:

$$f'(\pm\sqrt{r}) = r - 3(\sqrt{r})^2 = r - 3r = -2r.$$

Since $r > 0$, then $-2r < 0$. Thus, the two new branches are **stable**.

As r passes through 0, the fixed point at the origin loses stability and splits into two new stable fixed points. The bifurcation diagram is constructed by plotting the fixed point branches as explicit functions of the parameter r over the range $r \in [-2, 2]$, discretised with 400 equally spaced values. For $r \leq 0$, only the branch $x^* = 0$ exists and is stable, drawn as a blue line. For $r > 0$, this branch becomes unstable, drawn as a dashed red line, and two new stable branches $x^* = \pm\sqrt{r}$ emerge, also drawn as blue lines. The bifurcation point $r = 0$ is marked with a vertical grey line. We can observe the bifurcation diagram in Figure 2.3.

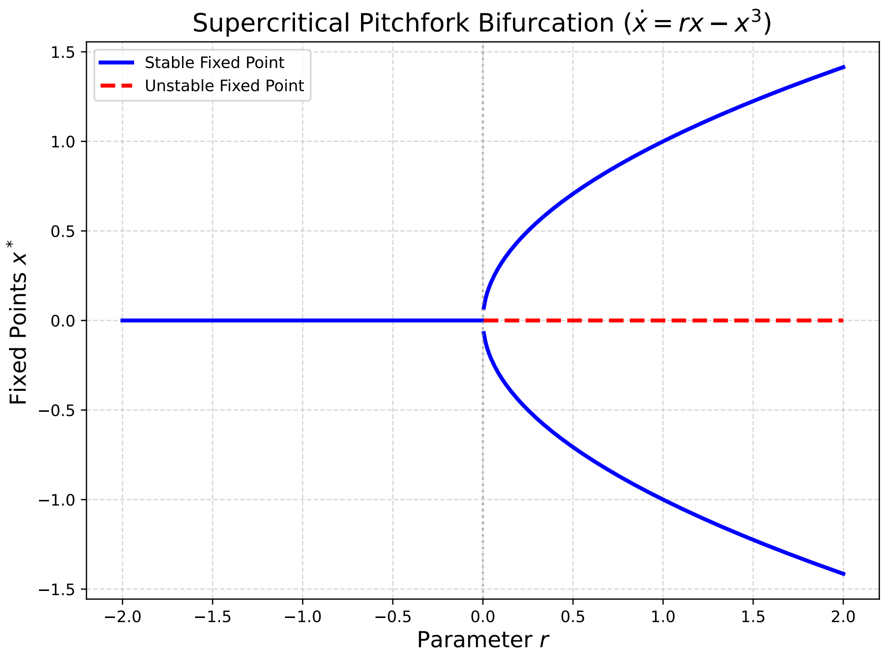

Fig. 2.3. Bifurcation diagram for the supercritical pitchfork bifurcation $\dot{x} = rx - x^3$.

2.2. Two-dimensional systems

Many natural phenomena involve multiple interacting variables. To observe richer dynamics, such as periodicity or chaos, we must look at higher dimensions.

A general two-dimensional system is given by:

$$\begin{cases} \dot{x} = f(x, y), \\ \dot{y} = g(x, y). \end{cases} \tag{2.9}$$

Here, the phase space is the Cartesian plane $\mathbb{R}^2$, and trajectories $(x(t), y(t))$ flow through this plane.

A linear system can be written in matrix form as:

$$\dot{\mathbf{x}} = A\mathbf{x}, \quad \text{where } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \text{ and } \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}. \tag{2.10}$$

The general solution is determined by the eigenvalues $\lambda_{1,2}$ and eigenvectors of the matrix A. The eigenvalues are the roots of the characteristic equation:

$$\det(A - \lambda I) = \lambda^2 - \tau\lambda + \Delta = 0,$$

where $\tau = \text{trace}(A) = a + d$ is the trace and $\Delta = \det(A) = ad - bc$ is the determinant. The eigenvalues are given by the quadratic formula:

$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}.$$

Following Strogatz [21], we can classify the fixed point at the origin $(0, 0)$ into five main topological types. The sign of Δ and the sign of the discriminant $(\tau^2 - 4\Delta)$ are the determining factors:

1. **Saddle Points** ($\Delta < 0$): The eigenvalues are real and have opposite signs. The fixed point is unstable. Trajectories approach the origin along one direction (the stable manifold, corresponding to the negative eigenvalue) and move away along another (the unstable manifold, corresponding to the positive eigenvalue).

2. **Nodes** ($\Delta > 0, \tau^2 - 4\Delta \geq 0$): The eigenvalues are real and have the same sign.

    - **Stable Node** ($\tau < 0$): Both eigenvalues are negative. All trajectories approach the origin monotonically (without oscillating).
    - **Unstable Node** ($\tau > 0$): Both eigenvalues are positive. All trajectories diverge from the origin.

3. **Spirals** ($\Delta > 0, \tau^2 - 4\Delta < 0$): The eigenvalues are complex conjugates $\alpha \pm i\omega$. The imaginary part implies rotation.

    - **Stable Spiral** ($\tau < 0$): The real part is negative. Trajectories spiral into the origin (damped oscillation).
    - **Unstable Spiral** ($\tau > 0$): The real part is positive. Trajectories spiral out to infinity (growing oscillation).

4. **Centers** ($\Delta > 0, \tau = 0$): The eigenvalues are purely imaginary ($\pm i\omega$). The trajectories are closed ellipses around the origin. This represents a conservative system with no energy loss or gain.

5. **Degenerate Nodes** ($\Delta = 0$ or $\tau^2 - 4\Delta = 0$): Borderline cases such as lines of fixed points or star nodes.

Example. To illustrate the phase plane analysis, let us revisit the linear system introduced previously:

$$\dot{x} = x + y, \quad \dot{y} = 4x - 2y.$$

The matrix is $A = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix}$. We compute the trace and determinant:

$$\tau = 1 + (-2) = -1, \quad \Delta = (1)(-2) - (1)(4) = -6.$$

Since $\Delta < 0$, the origin is a **saddle point**. The eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = -3$. The phase portrait is generated using the `streamplot` function from `Matplotlib`, which traces the flow lines of the vector field $(\dot{x}, \dot{y})$ across a uniform 100×100 grid over the domain $[-4, 4] \times [-4, 4]$. The unstable manifold, corresponding to the eigenvector $v_1 = (1, 1)$ associated with $\lambda_1 = 2$, is drawn as a dashed red line with slope 1. The stable manifold, corresponding to the eigenvector $v_2 = (1, -4)$ associated with $\lambda_2 = -3$, is drawn as a dash-dotted red line with slope -4.

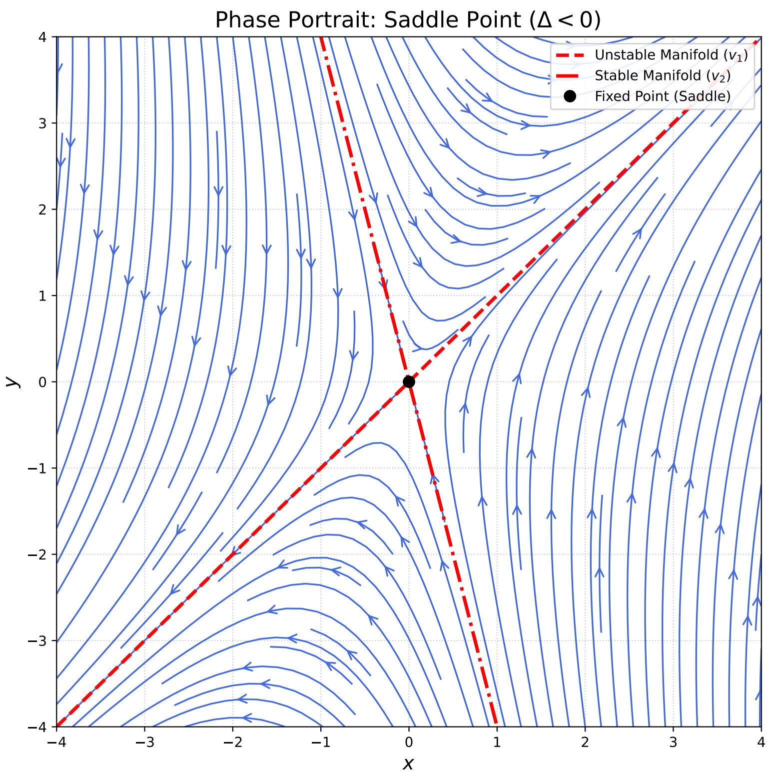

Fig. 2.4. Phase portrait for the linear system example

Linearisation

We can determine the local behaviour of trajectories near a fixed point by approximating the non-linear system with a linear one. This process is called *linearisation*.

Consider a general non-linear system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ with a fixed point $\mathbf{x}^*$. We consider a small perturbation $\boldsymbol{\eta}(t) = \mathbf{x}(t) - \mathbf{x}^*$. Using Taylor expansion and keeping only linear terms, the evolution of the perturbation is governed by:

$$\dot{\eta} = J\eta. \tag{2.11}$$

where J is the *Jacobian matrix* evaluated at the fixed point:

$$J(\mathbf{x}^*) = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix}_{\mathbf{x}=\mathbf{x}^*}.$$

According to the Hartman-Grobman Theorem [21], if all eigenvalues of the linearisation matrix J evaluated at $\mathbf{x}^*$ have non-zero real part (in which case the fixed point is called *hyperbolic*), then the phase portrait of the non-linear system near $\mathbf{x}^*$ is topologically equivalent to that of the linearised system $\dot{\boldsymbol{\eta}} = J\boldsymbol{\eta}$. Here, topological equivalence means that there exists a homeomorphism, that is, a continuous deformation with a continuous inverse, that maps trajectories onto trajectories while preserving the direction of time. In particular, the stability type of the fixed point (saddle, node, or spiral) is captured by the eigenvalues of J.

Example. Consider the Lotka-Volterra competition model (section 6.4 in [21]), where two species compete for the same food supply:

$$\begin{cases} \dot{x} = x(3 - x - 2y), \\ \dot{y} = y(2 - x - y). \end{cases} \tag{2.12}$$

where $x, y \geq 0$. To find the fixed points, we solve for $\dot{x} = 0$ and $\dot{y} = 0$. This yields four points: $(0, 0), (0, 2), (3, 0)$, and the coexistence point $(1, 1)$. Evaluating the Jacobian at the coexistence point $(1, 1)$:

$$J(1,1) = \begin{pmatrix} \frac{\partial \dot{x}}{\partial x} & \frac{\partial \dot{x}}{\partial y} \\ \frac{\partial \dot{y}}{\partial x} & \frac{\partial \dot{y}}{\partial y} \end{pmatrix}_{(1,1)} = \begin{pmatrix} 3 - 2x - 2y & -2x \\ -y & 2 - x - 2y \end{pmatrix}_{(1,1)} = \begin{pmatrix} -1 & -2 \\ -1 & -1 \end{pmatrix}.$$

Calculating the trace and determinant:

$$\tau = -2, \quad \Delta = (-1)(-1) - (-2)(-1) = 1 - 2 = -1.$$

Since $\Delta < 0$, the fixed point $(1, 1)$ is a **saddle point**. This means that coexistence is unstable; any small perturbation will lead the system away from $(1, 1)$, causing one species to drive the other to extinction. The phase portrait is generated via `streamplot` over a uniform 100×100 grid on the domain $[0, 3.5] \times [0, 2.5]$. The x-nullclines ($\dot{x} = 0$), given by $x = 0$ and $y = (3 - x)/2$, are drawn as blue dashed lines, and the y-nullclines ($\dot{y} = 0$), given by $y = 0$ and $y = 2 - x$, as green dashed lines. The saddle point $(1, 1)$ is marked in red, and the remaining fixed points $(0, 0)$, $(0, 2)$ and $(3, 0)$ in black.

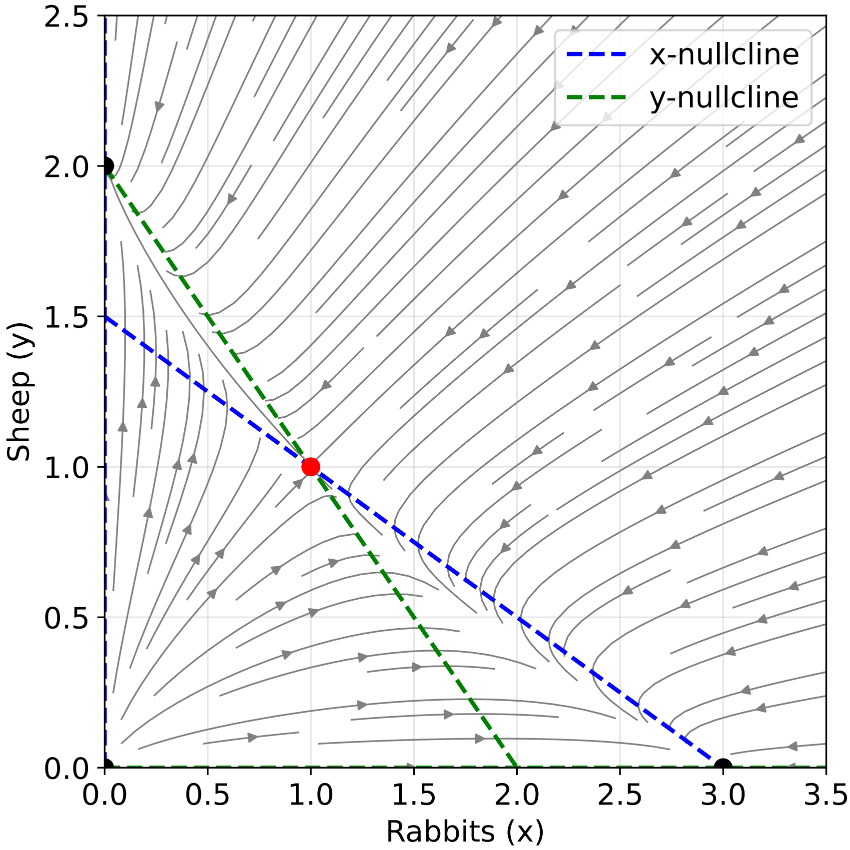

Fig. 2.5. Phase portrait for the rabbits vs. sheep model.

2.3. Limit cycles and the Poincaré-Bendixson Theorem

Non-linear systems can exhibit stable oscillations known as *limit cycles*.

Definition. A *limit cycle* is an isolated closed trajectory. Isolated means that neighboring trajectories are not closed; they spiral either towards (stable limit cycle) or away (unstable limit cycle) from the loop.

Before determining if a system possesses a limit cycle, it is useful to prove that closed orbits cannot exist in some systems. A powerful tool for this is the use of Lyapunov functions, which generalise the concept of "energy" or "dissipation" in a dynamical system.

Definition. Let $\mathbf{x}^*$ be a fixed point of a system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$. A *Lyapunov function* is a continuously differentiable, real-valued scalar function $V(\mathbf{x})$ defined on some domain D containing $\mathbf{x}^*$ such that:

1. $V(\mathbf{x}) > 0$ for all $\mathbf{x} \neq \mathbf{x}^*$ in D, and $V(\mathbf{x}^*) = 0$.

2. $\dot{V}(\mathbf{x}) < 0$ for all $\mathbf{x} \neq \mathbf{x}^*$ in D.

To evaluate the time derivative $\dot{V}$ along the trajectories of the system without explicitly solving the differential equations, we use the chain rule:

$$\dot{V} = \frac{d}{dt}V(\mathbf{x}) = \nabla V \cdot \dot{\mathbf{x}} = \nabla V \cdot \mathbf{f}(\mathbf{x}). \tag{2.13}$$

If such a function exists, the fixed point $\mathbf{x}^*$ is globally asymptotically stable within the domain D.

As a result, if a dynamical system admits a Lyapunov function $V(\mathbf{x})$ on a simply connected domain D, then the system has no closed trajectories lying entirely in D.

Since $V(\mathbf{x})$ must strictly decrease along any trajectory evolving in time ($\dot{V} < 0$), a trajectory can never return to its initial state. It is constantly losing energy, making persistent oscillations, such as limit cycles, impossible.

However, when a system is not purely dissipative and does not have a global Lyapunov function, trajectories might settle into a limit cycle.

Now we focus on the opposite question, how to determine when closed orbits exist in a certain system.

Theorem (Poincaré-Bendixson). Suppose that:

1. R is a closed, bounded subset of the plane.

2. $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ is a continuously differentiable vector field on an open set containing R.

3. R does not contain any fixed points.

4. There exists a trajectory that enters R and never leaves.

Then, that trajectory is either a closed orbit or approaches a closed orbit as $t \to \infty$.

Therefore, under the conditions of the Poincaré-Bendixson's Theorem, chaos is impossible in two-dimensional continuous systems, since trajectories are forced to eventually approach either a fixed point or a limit cycle.

In general, for a continuous deterministic system to exhibit chaos, the phase space must have at least three dimensions, which provides the motivation for the next chapter.

2.4. Bifurcations in higher dimensions

We have already introduced the pitchfork bifurcation for one-dimensional systems. Let us now introduce bifurcations in higher dimensions.

Bifurcations in higher dimensions can often be understood as a one-dimensional bifurcation embedded in a higher-dimensional phase space. For example, consider a supercritical pitchfork bifurcation in a two-dimensional system. The normal form can be extended as:

$$\begin{cases} \dot{x} = rx - x^3, \\ \dot{y} = -y. \end{cases} \tag{2.14}$$

Here, as the parameter r passes through zero, one eigenvalue of the Jacobian matrix crosses the imaginary axis from negative to positive (causing the bifurcation), while all other eigenvalues maintain negative real parts:

$$J(0,0) = \begin{pmatrix} r & 0 \\ 0 & -1 \end{pmatrix}, \tag{2.15}$$

which has eigenvalues $\lambda_1 = r$ and $\lambda_2 = -1$. The second eigenvalue $\lambda_2 = -1$ is always negative, while $\lambda_1 = r$ changes sign as r passes through zero: it is negative for $r < 0$, zero at $r = 0$, and positive for $r > 0$. Therefore, the origin loses stability exactly at $r = 0$, which is the bifurcation point.

Hopf Bifurcation

While one-dimensional bifurcations involve a single real eigenvalue passing through zero, two-dimensional systems allow for a different mechanism. Indeed, when a stable fixed point loses stability because a pair of complex conjugate eigenvalues crosses the imaginary axis, a *Hopf bifurcation* occurs.

Example. Consider the following non-linear system, where μ is a control parameter:

$$\begin{cases} \dot{x} = \mu x - y - x(x^2 + y^2), \\ \dot{y} = x + \mu y - y(x^2 + y^2). \end{cases} \tag{2.16}$$

First, we analyse the linear stability of the origin $(0,0)$. The Jacobian matrix evaluated at the origin is:

$$J(0,0) = \begin{pmatrix} \mu & -1 \\ 1 & \mu \end{pmatrix}$$

We compute the trace τ and determinant Δ:

$$\tau = 2\mu, \quad \Delta = \mu^2 + 1.$$

The eigenvalues are given by $\lambda = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}$. Substituting our values yields:

$$\lambda_{1,2} = \mu \pm i.$$

Notice what happens as the parameter μ varies:

- When $\mu < 0$, the eigenvalues have a negative real part. The origin is a **stable spiral**.
- When $\mu = 0$, the eigenvalues are purely imaginary ($\pm i$). The real part crosses the imaginary axis.
- When $\mu > 0$, the eigenvalues have a positive real part. The origin becomes an **unstable spiral**.

This crossing of complex conjugate eigenvalues is the Hopf bifurcation. To understand what happens to the trajectories when the origin becomes unstable ($\mu > 0$), it is convenient to rewrite the system in polar coordinates.

Using the transformations $r^2 = x^2 + y^2$ and $\tan(\theta) = y/x$, we can differentiate with respect to time to obtain $r\dot{r} = x\dot{x} + y\dot{y}$ and $\dot{\theta} = (x\dot{y} - y\dot{x})/r^2$. Substituting the original equations into these expressions yields the uncoupled system:

$$\begin{cases} \dot{r} = \mu r - r^3, \\ \dot{\theta} = 1. \end{cases} \tag{2.17}$$

Now we analyse the radial dynamics by solving $\dot{r} = 0$:

$$r(\mu - r^2) = 0.$$

1. Case $\mu < 0$: The only real root is $r^* = 0$. For any $r > 0$, we have $\dot{r} < 0$. All trajectories spiral inwards towards the origin, confirming it is a stable spiral.
2. Case $\mu > 0$: The origin $r^* = 0$ is still a fixed point, but since $\dot{r} > 0$ for small r, it is an unstable spiral. A new steady state emerges at $r^* = \sqrt{\mu}$. Evaluating the flow around this new radius: if $r < \sqrt{\mu}$, then $\dot{r} > 0$ (trajectories move outward); if $r > \sqrt{\mu}$, then $\dot{r} < 0$ (trajectories move inward).

Therefore, for $\mu > 0$, the system exhibits a **stable limit cycle** of radius $\sqrt{\mu}$. As μ passes through zero, the stable resting state transitions into an oscillating state. This is an example of a *supercritical Hopf bifurcation*.

These concepts are fundamental for analysing chaotic systems, such as the Lorenz system.

3. CHAOS THEORY

3.1. The Lorenz system

Edward Lorenz wrote in 1963 an important paper in which he derived a system of differential equations that model convection motion in a two-dimensional fluid cell; for further information, see [21], [20] and [14].

For different values of the parameters, to approximate solutions for the equations was a difficult task, since for some parameter values, numerically computed solutions appeared to show random, oscillating behaviour. This was the first time that chaotic behaviour was observed numerically. For this reason, these equations became of interest for different scientists that tried to relate this behaviour with other dynamical systems.

3.1.1. Lorenz equations

The Lorenz equations can be written as the following system of ordinary differential equations:

$$\begin{cases} \dot{x} = \sigma(y - x), \\ \dot{y} = rx - y - xz, \\ \dot{z} = xy - bz. \end{cases} \tag{3.1}$$

The variable x measures the rate of convective overturning, the variable y measures the horizontal temperature variation, and the variable z measures the vertical temperature variation. The three parameters σ, r, and b are respectively proportional to the Prandtl number, the Rayleigh number, and some physical proportions of the region under consideration. Consequently, all three are taken to be positive.

More precisely, x is proportional to the intensity of the convective motion, y is proportional to the temperature difference between the ascending and descending currents, with similar signs of x and y denoting that warm fluid is rising and cold fluid is descending. Finally, the variable z is proportional to the distortion of the vertical temperature profile from linearity. A positive value indicates that the strongest temperature gradients occur near the boundaries of the fluid cell.

The Lorenz equations exhibit several properties that are essential for understanding their complex dynamics.

Symmetry

If we replace the variables (x, y, z) with $(-x, -y, z)$, the system of equations remains unchanged.

Let us define new variables $u = -x$, $v = -y$, and $w = z$. Taking the time derivative of these new variables yields:

$$\begin{cases} \dot{u} = -\dot{x}, \\ \dot{v} = -\dot{y}, \\ \dot{w} = \dot{z}. \end{cases} \tag{3.2}$$

Substituting the original Lorenz equations into these derivatives, we obtain:

$$\begin{cases} \dot{u} = -\sigma(y - x) = \sigma(-y - (-x)) = \sigma(v - u), \\ \dot{v} = -(rx - y - xz) = r(-x) - (-y) - (-x)z = ru - v - uw, \\ \dot{w} = xy - bz = (-x)(-y) - bz = uv - bw. \end{cases} \tag{3.3}$$

This implies that if $(x(t), y(t), z(t))$ is a valid solution, then $(-x(t), -y(t), z(t))$ is also a valid solution. From the physical perspective, this symmetry reflects the fact that the convective fluid can circulate either clockwise or counterclockwise with equal probability.

Volume contraction and dissipation

One of the most important features of the Lorenz system is that it is dissipative. A system is called dissipative if arbitrary volumes in its phase space contract and shrink to zero as time evolves. Following [14], we can calculate the divergence of the vector field $\mathbf{f}(x, y, z) = (\dot{x}, \dot{y}, \dot{z})$:

$$\nabla \cdot \mathbf{f} = \frac{\partial \dot{x}}{\partial x} + \frac{\partial \dot{y}}{\partial y} + \frac{\partial \dot{z}}{\partial z} = -\sigma - 1 - b. \tag{3.4}$$

Since the parameters σ and b are strictly positive, the divergence is constant and strictly negative. This means that an arbitrary volume V in the phase space contracts in time according to the equation $\dot{V} = -(\sigma + 1 + b)V$.

The solution to this differential equation is $V(t) = V(0)e^{-(\sigma+1+b)t}$. Therefore, as $t \to \infty$, the volume shrinks to zero. This implies that all trajectories converge to a subset of phase space of zero volume, which will eventually contain the strange attractor (will be defined in next sections).

Fixed points and stability

The fixed points of the system are found by setting the derivatives to zero ($\dot{x} = \dot{y} = \dot{z} = 0$):

$$\sigma(y - x) = 0, \tag{3.5}$$

$$rx - y - xz = 0, \tag{3.6}$$

$$xy - bz = 0. \tag{3.7}$$

From Equation (3.5), we deduce that $y = x$. Substituting this into Equation (3.7) yields $z = x^2/b$. Finally, plugging both expressions into Equation (3.6) provides $x(r-1-x^2/b) = 0$. This polynomial yields different fixed points depending on the value of the parameter r:

- If $r < 1$: The term $(r - 1 - x^2/b) = 0$ implies $x^2 = b(r - 1)$. Since x must be a real number and $b > 0$, this equation has no real solutions. Therefore, the origin $C_0 = (0, 0, 0)$ is the only fixed point of the system. It corresponds physically to a state of no convection, where heat is transported entirely by conduction.

- If $r = 1$: The equation yields $x^2 = 0$, meaning the origin $C_0 = (0, 0, 0)$ is still the only fixed point. However, this is a critical parameter value where the system undergoes a *supercritical pitchfork bifurcation*.

- If $r > 1$: The origin C_0 still exists, but the term $(r - 1 - x^2/b) = 0$ now yields two additional real roots. This results in the creation of a pair of symmetric fixed points:

  $$C^{\pm} = \left(\pm\sqrt{b(r-1)}, \pm\sqrt{b(r-1)}, r-1\right). \tag{3.8}$$

  Physically, these points represent steady convective rolls rotating in opposite directions.

Let us analyse the stability of these fixed points as the control parameter r increases.

First, we consider the regime where $0 < r < 1$. In this scenario, we must analyse the stability of the origin $C_0 = (0, 0, 0)$. We do this in two steps: first by studying its local linear stability, and then by proving its global stability using a Lyapunov function, following the approach outlined in [21].

We linearize the system around the fixed point $(0, 0, 0)$ by computing the Jacobian matrix $J(x, y, z)$:

$$J(x,y,z) = \begin{pmatrix} -\sigma & \sigma & 0 \\ r-z & -1 & -x \\ y & x & -b \end{pmatrix}. \tag{3.9}$$

Evaluating the Jacobian at the origin yields:

$$J(0,0,0) = \begin{pmatrix} -\sigma & \sigma & 0 \\ r & -1 & 0 \\ 0 & 0 & -b \end{pmatrix}. \tag{3.10}$$

To find the eigenvalues λ, we solve the characteristic equation $\det(J - \lambda I) = 0$. Because the third column has only one non-zero entry on the diagonal, we immediately obtain one eigenvalue in the z-direction:

$$\lambda_3 = -b. \tag{3.11}$$

Since $b > 0$, this eigenvalue is strictly negative, indicating exponential decay along the z-axis. The remaining two eigenvalues are determined by:

$$\det\begin{pmatrix} -\sigma - \lambda & \sigma \\ r & -1 - \lambda \end{pmatrix} = 0. \tag{3.12}$$

This gives the quadratic equation $\lambda^2 + (\sigma+1)\lambda + \sigma(1-r) = 0$. Using the quadratic formula, the roots are:

$$\lambda_{1,2} = \frac{-(\sigma + 1) \pm \sqrt{(\sigma + 1)^2 - 4\sigma(1 - r)}}{2}. \tag{3.13}$$

For $r < 1$, the term $1 - r$ is strictly positive. Consequently, the term $4\sigma(1 - r)$ is positive, which means $\sqrt{(\sigma + 1)^2 - 4\sigma(1 - r)} < (\sigma + 1)$. Therefore, both eigenvalues λ_1 and λ_2 are real and strictly negative. Since all three eigenvalues $(\lambda_1, \lambda_2, \lambda_3)$ are negative, the origin is locally asymptotically stable for $r < 1$.

Linear stability only guarantees that trajectories starting near the origin will converge to it. To prove that all trajectories, regardless of their initial conditions, eventually decay to the origin when $r < 1$, we construct a Lyapunov function.

Consider the following scalar function $V(x, y, z)$:

$$V(x, y, z) = \frac{1}{\sigma}x^2 + y^2 + z^2. \tag{3.14}$$

Notice that $V(x, y, z) \geq 0$ for all (x, y, z), and $V(x, y, z) = 0$ if and only if $x = y = z = 0$. To prove global stability, we must show that the time derivative $\dot{V}$ is strictly negative everywhere except at the origin. Taking the derivative with respect to time and substituting the Lorenz equations yields:

$$\begin{aligned}
\frac{1}{2}\dot{V} &= \frac{1}{\sigma}x\dot{x} + y\dot{y} + z\dot{z} \\
&= x(y - x) + y(rx - y - xz) + z(xy - bz) \\
&= xy - x^2 + rxy - y^2 - xyz + xyz - bz^2 \\
&= (r + 1)xy - x^2 - y^2 - bz^2
\end{aligned} \tag{3.15}$$

To determine the sign of this expression, we complete the square for the terms involving x and y:

$$\frac{1}{2}\dot{V} = -\left(x - \frac{r + 1}{2}y\right)^2 - \left[1 - \left(\frac{r + 1}{2}\right)^2\right]y^2 - bz^2. \tag{3.16}$$

For $\dot{V}$ to be strictly negative for all $(x, y, z) \neq (0, 0, 0)$, the coefficient of y^2 must be positive. That requires:

$$1 - \left(\frac{r + 1}{2}\right)^2 > 0 \implies (r + 1)^2 < 4 \implies r + 1 < 2 \implies r < 1 \tag{3.17}$$

(assuming $r > 0$). Thus, if $r < 1$, $\dot{V} < 0$ everywhere except at the origin.

By Lyapunov's method, this proves that for $r < 1$, the origin is a globally stable fixed point.

As we increase the heating, at exactly $r = 1$, the system undergoes a *supercritical pitchfork bifurcation*. The origin loses its stability, and the two new stable fixed points C^+ and C^- are created.

For parameter values $r > 1$, the origin acts as an unstable saddle, and the system behaviour is instead governed by the convective rolls $C^\pm = \left(\pm\sqrt{b(r-1)}, \pm\sqrt{b(r-1)}, r-1\right)$. To analyse their local stability, we evaluate the Jacobian matrix at these coordinates:

$$J(C^\pm) = \begin{pmatrix} -\sigma & \sigma & 0 \\ 1 & -1 & \mp\sqrt{b(r-1)} \\ \pm\sqrt{b(r-1)} & \pm\sqrt{b(r-1)} & -b \end{pmatrix}. \tag{3.18}$$

By solving the characteristic equation $\det(J - \lambda I) = 0$, we obtain the following cubic polynomial for the eigenvalues:

$$\lambda^3 + (\sigma + b + 1)\lambda^2 + b(r + \sigma)\lambda + 2\sigma b(r - 1) = 0. \tag{3.19}$$

Let us define the coefficients of this cubic polynomial as $A = \sigma + b + 1$, $B = b(r + \sigma)$, and $C = 2\sigma b(r - 1)$. Thus, the characteristic equation is $\lambda^3 + A\lambda^2 + B\lambda + C = 0$. Since we are analysing the domain where $r > 1$, and the physical parameters σ and b are strictly positive, it follows that $A > 0$, $B > 0$, and $C > 0$.

We are interested in the parameter value where the fixed points lose their stability. This loss of stability occurs exactly when a pair of complex conjugate eigenvalues cross the imaginary axis in the complex plane, meaning their real part becomes zero. This is when the *Hopf bifurcation* appears.

To find the critical Rayleigh number, denoted as r_H, we assume a purely imaginary root $\lambda = i\omega$ (with $\omega \neq 0$) and substitute it into the characteristic equation:

$$(i\omega)^3 + A(i\omega)^2 + B(i\omega) + C = 0 \implies -i\omega^3 - A\omega^2 + iB\omega + C = 0. \tag{3.20}$$

By separating the real and imaginary parts, we get a system of two equations. For the real part, $C - A\omega^2 = 0 \implies \omega^2 = \frac{C}{A}$. For the imaginary part, $B\omega - \omega^3 = 0 \implies \omega(B - \omega^2) = 0 \implies \omega^2 = B$.

Equating both expressions for ω^2 yields the condition $C = AB$. Now, we substitute our original parameter definitions back into this condition to solve for r_H:

$$2\sigma b(r - 1) = (\sigma + b + 1)b(r + \sigma). \tag{3.21}$$

Since $b > 0$, we can divide both sides by b:

$$2\sigma(r-1) = (\sigma+b+1)(r+\sigma). \tag{3.22}$$

Expanding both sides:

$$2\sigma r - 2\sigma = (\sigma+b+1)r + \sigma(\sigma+b+1). \tag{3.23}$$

Next, we group all terms involving r on the left side:

$$2\sigma r - (\sigma+b+1)r = \sigma(\sigma+b+1) + 2\sigma. \tag{3.24}$$

Factoring out r:

$$r(2\sigma - \sigma - b - 1) = \sigma^2 + \sigma b + \sigma + 2\sigma. \tag{3.25}$$

Simplifying the expressions inside the parentheses:

$$r(\sigma - b - 1) = \sigma(\sigma + b + 3). \tag{3.26}$$

Finally, isolating r, we obtain the critical Rayleigh number for the Hopf bifurcation:

$$r_H = \sigma\left(\frac{\sigma+b+3}{\sigma-b-1}\right). \tag{3.27}$$

Since the physical parameters σ and b are strictly positive, the numerator $(\sigma + b + 3)$ is always positive. For r_H to exist in the physical domain ($r > 0$), the denominator must also be positive, leading directly to the condition $\sigma - b - 1 > 0$. Thus, this critical value is only physically meaningful if the condition $\sigma > b + 1$ is satisfied.

For the classical Lorenz parameters ($\sigma = 10$ and $b = 8/3$), the Hopf bifurcation occurs at exactly:

$$r_H = \sigma\left(\frac{\sigma+b+3}{\sigma-b-1}\right) = 10\left(\frac{10+\frac{8}{3}+3}{10-\frac{8}{3}-1}\right) = 10\left(\frac{\frac{47}{3}}{\frac{19}{3}}\right) = \frac{470}{19} \approx 24.74. \tag{3.28}$$

For $1 < r < r_H$, the fixed points C^+ and C^- are stable, acting as spiral sinks where trajectories slowly converge. As r crosses r_H, a subcritical Hopf bifurcation occurs. The fixed points become unstable spiral sources, repelling nearby trajectories and entering the chaotic region that define the strange attractor.

3.2. Defining deterministic chaos

Although there is no universally accepted mathematical definition for chaos, mathematicians and physicists generally agree on its fundamental properties. Following [21], chaos can be defined as *aperiodic long-term behaviour in a deterministic system that exhibits sensitive dependence on initial conditions*.

There are three main characteristics in this definition:

- **Deterministic:** The system has no random inputs or parameters.
- **Aperiodic long-term behaviour:** This means that as time $t \to \infty$, the system's trajectories do not settle down to fixed points or periodic orbits. Instead, they continue to oscillate irregularly.
- **Sensitive dependence on initial conditions:** This implies that two almost identical initial states will diverge from each other, making long-term prediction impossible.

Remark. Moreover, it can also be defined from a topological point of view too, established by Devaney [5]. However, for simplicity we will not focus on this approach.

3.2.1. Sensitivity to initial conditions

The most remarkable feature of chaotic systems is their *sensitive dependence on the initial conditions*. This means that even if a system is deterministic, any infinitesimal error or uncertainty in the measurement of the initial state will result in extremely different results over time. This phenomenon is popularly known as the *butterfly effect*.

In order to analyse this property, we introduce the concept of the **Lyapunov exponent**. Suppose we have a dynamical system and we consider two initial states that are extremely close to each other. Let the first state be $\mathbf{x}_0$ and the second state be $\mathbf{x}_0 + \delta\mathbf{x}_0$, where the initial separation vector $\delta\mathbf{x}_0$ has a very small magnitude $||\delta\mathbf{x}_0||$.

Let $\delta\mathbf{x}(t)$ represent the separation between the two trajectories after time t. In a chaotic system, this separation grows exponentially fast, which can be expressed as:

$$||\delta\mathbf{x}(t)|| \approx ||\delta\mathbf{x}_0||e^{\lambda t}, \tag{3.29}$$

where λ is the Lyapunov exponent [21].

By rearranging this equation and taking the limit for very small initial perturbations and very long times, the (maximal) Lyapunov exponent can be defined as:

$$\lambda = \lim_{t\to\infty} \lim_{||\delta\mathbf{x}_0||\to 0} \frac{1}{t} \ln\left(\frac{||\delta\mathbf{x}(t)||}{||\delta\mathbf{x}_0||}\right). \tag{3.30}$$

The sign of λ indicates the stability of the system's dynamics:

- If $\lambda < 0$, nearby trajectories converge. This is characteristic of systems that settle into stable fixed points or stable limit cycles. The system is not sensitive to initial conditions and is highly predictable.

- If $\lambda = 0$, the trajectories maintain a constant separation. This behaviour is typical of marginally stable systems or conservative systems without dissipation.

- If $\lambda > 0$, nearby trajectories diverge exponentially. This positive exponent represents that there is sensitive dependence on initial conditions and is a fundamental requirement for chaos.

In an n-dimensional phase space, a small initial sphere of perturbed initial conditions will deform into an n-dimensional ellipsoid as the system evolves. Therefore, an n-dimensional system has n different Lyapunov exponents $\{\lambda_1, \lambda_2, \ldots, \lambda_n\}$, each corresponding to the rate of expansion or contraction along one of the principal axes of the ellipsoid.

Usually, they are ordered from largest to smallest $\{\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n\}$. The largest of these, λ_1, is called the *Maximal Lyapunov Exponent* (MLE). If $\lambda_1 > 0$, the dominant behaviour is exponential divergence, and the system is chaotic, regardless of the values of the other exponents.

For the Lorenz system, we can deduce the exact signs of its three Lyapunov exponents $\{\lambda_1, \lambda_2, \lambda_3\}$ simply by combining the definition of chaos with the system's properties previously analysed:

1. For the MLE (λ_1): The definition of a chaos requires sensitive dependence on initial conditions. Therefore, at least one direction must exhibit exponential divergence, meaning that the maximal Lyapunov exponent must be strictly positive: $\lambda_1 > 0$.

2. For λ_2: Since the chaotic trajectory never settles to a fixed point and is bounded, this longitudinal separation neither grows exponentially to infinity nor shrinks to zero. Consequently, for any continuous-time system lacking a stable equilibrium, the Lyapunov exponent corresponding to the flow direction is exactly zero: $\lambda_2 = 0$.

3. For λ_3: By Liouville's theorem, the average exponential rate of phase space volume evolution is governed by the sum of all Lyapunov exponents [18]:

$$\sum_{i=1}^{n} \lambda_i = \lim_{t\to\infty} \frac{1}{t} \int_0^t (\nabla \cdot \mathbf{f}(\mathbf{x}(\tau)))\, d\tau. \tag{3.31}$$

As we previously proved in the volume contraction analysis, the divergence of the Lorenz vector field is strictly constant and negative everywhere: $\nabla \cdot \mathbf{f} = -(\sigma + 1 + b)$. Therefore, the sum of the Lyapunov exponents must equal this constant divergence:

$$\lambda_1 + \lambda_2 + \lambda_3 = -(\sigma + 1 + b). \tag{3.32}$$

Since $\lambda_2 = 0$ and $\lambda_1 > 0$, the third exponent must be strongly negative to satisfy the equation:

$$\lambda_3 = -(\sigma + 1 + b) - \lambda_1. \tag{3.33}$$

Lyapunov exponent for one-dimensional maps

One-dimensional maps provide one of the simplest systems to derive Lyapunov exponents [18, 21].

Consider a one-dimensional iterative map defined by the difference equation:

$$x_{n+1} = f(x_n), \tag{3.34}$$

where n represents the discrete time step. Suppose we have an initial condition x_0 and a nearby perturbed state $x_0 + \delta x_0$. After one iteration, the separation between the two orbits is:

$$\delta x_1 = f(x_0 + \delta x_0) - f(x_0). \tag{3.35}$$

Assuming the initial perturbation δx_0 is infinitesimally small, we can approximate this difference using a Taylor expansion, keeping only the linear term:

$$\delta x_1 \approx f'(x_0)\delta x_0. \tag{3.36}$$

If we iterate this process n times, the separation at time step n becomes:

$$\delta x_n \approx (f^n)'(x_0)\delta x_0, \tag{3.37}$$

where $f^n(x_0)$ denotes the n-th composition of the map f.

To understand how we evaluate the derivative $(f^n)'(x_0)$, we must apply the chain rule. Consider the second iterate, $f^2(x_0) = f(f(x_0))$. By applying the chain rule, we obtain:

$$(f^2)'(x_0) = \frac{d}{dx_0}[f(f(x_0))] = f'(f(x_0)) \cdot f'(x_0) = f'(x_1)f'(x_0). \tag{3.38}$$

Extending this process to n iterations:

$$(f^n)'(x_0) = f'(x_{n-1})f'(x_{n-2})\dots f'(x_1)f'(x_0) = \prod_{i=0}^{n-1} f'(x_i). \tag{3.39}$$

By the definition of sensitive dependence on initial conditions, we assume that the magnitude of the separation grows exponentially with n, such that $|\delta x_n| \approx |\delta x_0|e^{\lambda n}$. Equating this exponential growth to our chain-rule expansion yields:

$$|\delta x_0|e^{\lambda n} \approx |\delta x_0| \left| \prod_{i=0}^{n-1} f'(x_i) \right|. \tag{3.40}$$

Canceling $|\delta x_0|$ from both sides, taking the natural logarithm, and dividing by n, we obtain an expression for λ:

$$\lambda \approx \frac{1}{n} \ln \left| \prod_{i=0}^{n-1} f'(x_i) \right| = \frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x_i)|. \tag{3.41}$$

Finally, taking the limit as $n \to \infty$ gives the formal definition of the Lyapunov exponent for a one-dimensional discrete map:

$$\lambda = \lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x_i)|. \tag{3.42}$$

If this value is positive ($\lambda > 0$), the map exhibits sensitive dependence on initial conditions and is defined as chaotic. This formula will be fundamental for the analysis of the logistic map in the upcoming sections.

Maximum Lyapunov Exponent computation for higher-dimensional systems

The Lyapunov exponent formula derived in the previous section applies exclusively to one-dimensional discrete maps, where the linearisation reduces to a scalar derivative. For systems defined on $\mathbb{R}^n$ with $n > 1$, the growth of a perturbation depends on its direction, and in general n distinct exponents exist. However, as stated in the previous section, the sign of the largest of them, the Maximal Lyapunov Exponent λ_1, is both necessary and sufficient to determine whether a system exhibits sensitive dependence on initial conditions. Therefore, it is the only exponent needed to detect chaos, and it can be estimated numerically.

Given a reference trajectory $\mathbf{x}(t)$ and a shadow trajectory starting at distance δ_0 from it, both are integrated forward for a short time interval τ. The separation between them after each interval,

$$d_k = \|\mathbf{x}_{\text{shad}}(k\tau) - \mathbf{x}_{\text{ref}}(k\tau)\|, \tag{3.43}$$

is measured before the shadow trajectory is rescaled back to distance δ_0 from the reference in the same direction. Accumulating the logarithmic growth over N such steps yields the estimate

$$\lambda_1 \approx \frac{1}{N\tau} \sum_{k=1}^{N} \ln \frac{d_k}{\delta_0}. \tag{3.44}$$

The periodic rescaling prevents the two trajectories from leaving the linear regime while the accumulated logarithms track the average exponential growth rate. The sign of λ_1 then provides the criterion already stated: the system is chaotic if $\lambda_1 > 0$ and stable if $\lambda_1 \leq 0$.

Remark. The generalisation of this procedure to compute the full spectrum $\lambda_1 \geq \cdots \geq \lambda_n$ requires tracking n independent perturbation vectors simultaneously and applying Gram–Schmidt orthonormalisation at each rescaling step to prevent their collapse onto the dominant direction. This is the algorithm developed in [2]. However, for the purposes of this report, we will only focus on the MLE.

3.3. The logistic map

The logistic map is the discretise version of the logistic equation, proposed by Robert May, mainly used to measure population growth:

$$x_{n+1} = rx_n(1 - x_n) \tag{3.45}$$

where $x_n \in [0, 1]$ represents the population at generation n (normalized to the maximum carrying capacity of the environment), and r represents the growth rate.

We will analyse this map in depth in several ways in order to visualize the different behaviours of it regarding parameter r. No negative values of this parameter will be considered, as it would result in negative population values.

Moreover, the function $f(x) = rx(1-x)$ is a parabola with its maximum located at $x = 1/2$. Consequently, the maximum height the function can reach is $f(1/2) = r/4$.

Because x_n represents a population fraction, the map must restrict the output within the interval $[0, 1]$. If the peak of the parabola were to exceed 1, following iterations could escape the unit interval and diverge to $-\infty$. Therefore, we require the maximum height to be $r/4 \leq 1$, which bounds the parameter to the range $r \in [0, 4]$.

3.3.1. Analysis of the logistic map

Fixed points and stability

Setting $x_{n+1} = x_n = x^*$, we obtain:

$$x^* = rx^*(1 - x^*),$$

$$rx^*(1 - x^*) - x^* = 0,$$

$$x^*[r(1 - x^*) - 1] = 0.$$

From this expression we get two different fixed points: $x_1^* = 0$ and $x_2^* = 1 - \frac{1}{r}$.

To determine the linear stability of these points, we evaluate the multiplier λ, $f'(x^*) = r(1 - 2x^*)$. A fixed point is stable if $|\lambda| < 1$ and unstable if $|\lambda| > 1$.

For the origin, we get $f'(0) = r$. Each of the following figures consists of two panels: the left panel shows the time series x_n as a function of the generation n, and the right panel shows the corresponding cobweb diagram. In all cases, the initial condition is fixed at $x_0 = 0.1$ and the map is iterated for $n = 50$ steps, which is sufficient to reveal the asymptotic behaviour of the system. The parabola $f(x) = rx(1 - x)$ and the diagonal $y = x$ are evaluated on a uniform grid of 400 points over $[0, 1]$. The cobweb path is traced following the algorithm described in Chapter 2. We can show how the population goes to extinction for $r = 0.5$ in the following cobweb diagram:

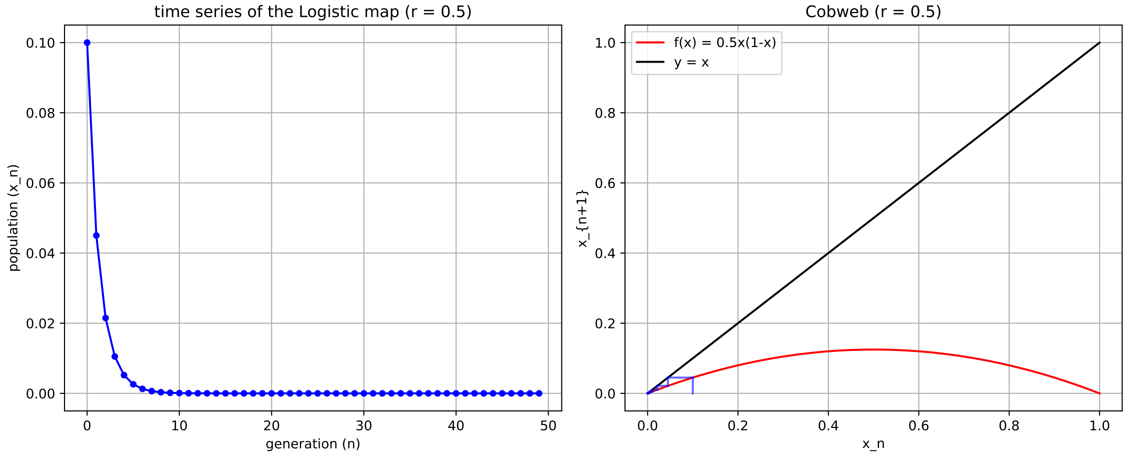

Fig. 3.1. Cobweb diagram for the logistic map with $r = 0.5$.

At $r = 1$, a *transcritical bifurcation* occurs, and the origin becomes unstable for $r > 1$. Now, evaluating the derivative for $x_2^* = 1 - 1/r$:

$$\lambda = f'\left(1 - \frac{1}{r}\right) = r\left[1 - 2\left(1 - \frac{1}{r}\right)\right] = r\left(1 - 2 + \frac{2}{r}\right) = 2 - r. \qquad (3.46)$$

In order to be stable, it must satisfy $|2 - r| < 1$, which is the same as satisfying $-1 < 2 - r < 1$, ending with the condition $1 < r < 3$. Therefore, for growth rates between 1 and 3, the population tends to stabilise.

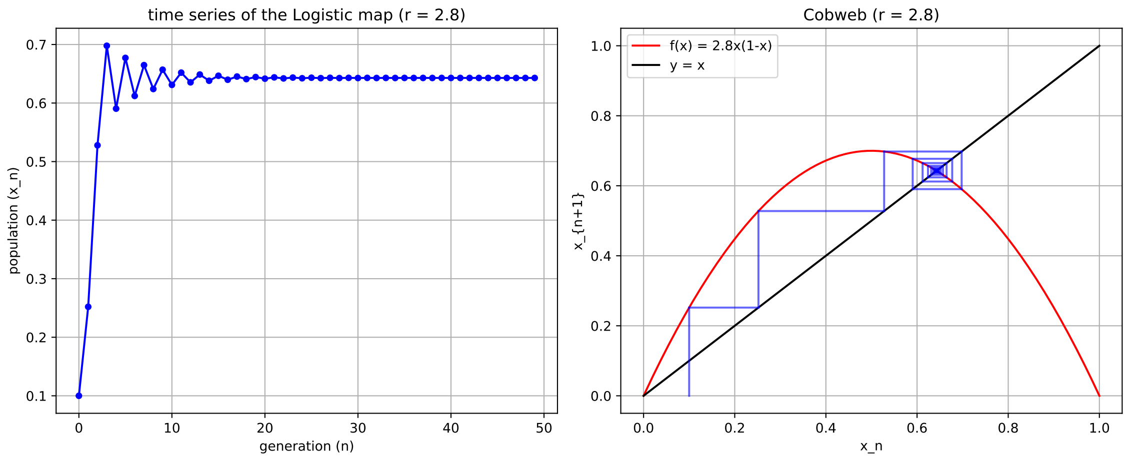

Fig. 3.2. Cobweb diagram for the logistic map with $r = 2.8$.

We will see now how stability changes for $r = 3$.

Period-doubling bifurcation

Next, (see[21] , section 10.3), we can prove that at $r = 3$, the fixed point loses stability, and a new stable cycle of period 2 appears. This is known as a *period-doubling* or flip bifurcation. The population begins to oscillate between two different values, p and q, such that $f(p) = q$ and $f(q) = p$. This is equivalent to have $f(f(p)) = p$, which is a fixed point of the second iterate map.

In order to prove this, we need to solve $f(f(x)) = x$.

$$f(f(x)) = r[rx(1-x)][1-rx(1-x)] = x. \tag{3.47}$$

Since the fixed points of $f(x)$ ($x_1^* = 0$ and $x_2^* = 1 - \frac{1}{r}$) are also fixed points of $f^2(x)$, we can factor out x and $(x - (1 - \frac{1}{r}))$. Dividing the polynomial by these roots yields to the following polynomial:

$$r^3x^2 - r^2(r+1)x + r(r+1) = 0, \tag{3.48}$$

Thus, we obtain

$$x_{p,q} = \frac{r + 1 \pm \sqrt{(r+1)(r-3)}}{2r}. \tag{3.49}$$

Finally, these values only exist when $r \geq 3$, demonstrating that at this value a period-doubling bifurcation occurs.

Moreover, we can determine the stability of the 2-cycle.

To do so, we compute the multiplier for $f^2(x)$ using the chain rule:

$$\lambda = (f^2)'(p) = f'(f(p))f'(p) = f'(q)f'(p). \tag{3.50}$$

Substituting $f'(x) = r(1-2x)$ and evaluating at the roots p and q, we obtain:

$$\lambda = r(1-2p) \cdot r(1-2q) = r^2[1 - 2(p+q) + 4pq]. \tag{3.51}$$

The sum of the roots is $p + q = \frac{r+1}{r}$ and the product is $pq = \frac{r+1}{r^2}$. Substituting these values:

$$\lambda = r^2\left[1 - 2\left(\frac{r+1}{r}\right) + 4\left(\frac{r+1}{r^2}\right)\right] = 4 + 2r - r^2. \tag{3.52}$$

The 2-cycle is stable when $|4 + 2r - r^2| < 1$. Solving this, we find that the 2-cycle is stable at $3 < r < 1 + \sqrt{6} \approx 3.449$. At this point, another period-doubling bifurcation occurs, giving birth to a stable 4-cycle.

We can observe these results graphically in the following cobweb diagrams, for $r = 3.3$, $r = 3.8$ and $r = 4$ respectively:

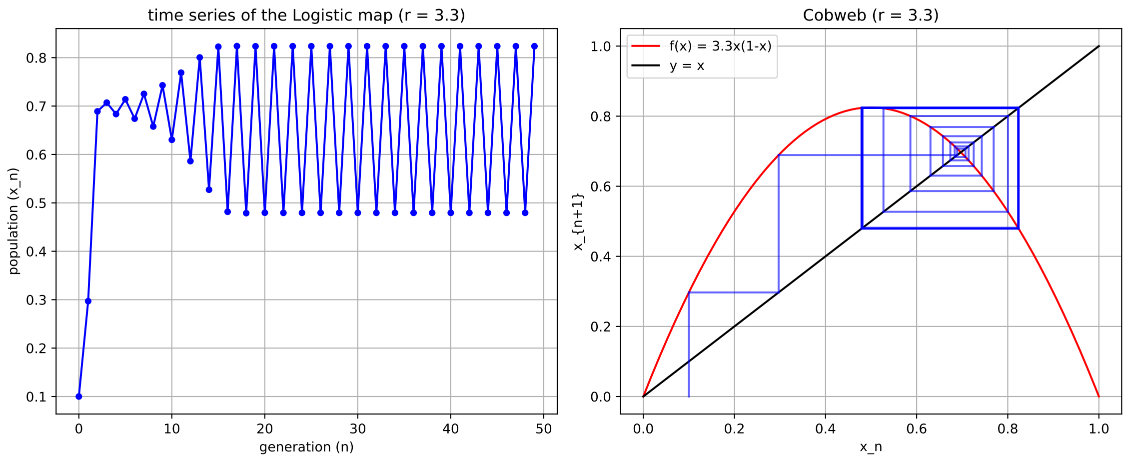

Fig. 3.3. Cobweb diagram for the logistic map with $r = 3.3$.

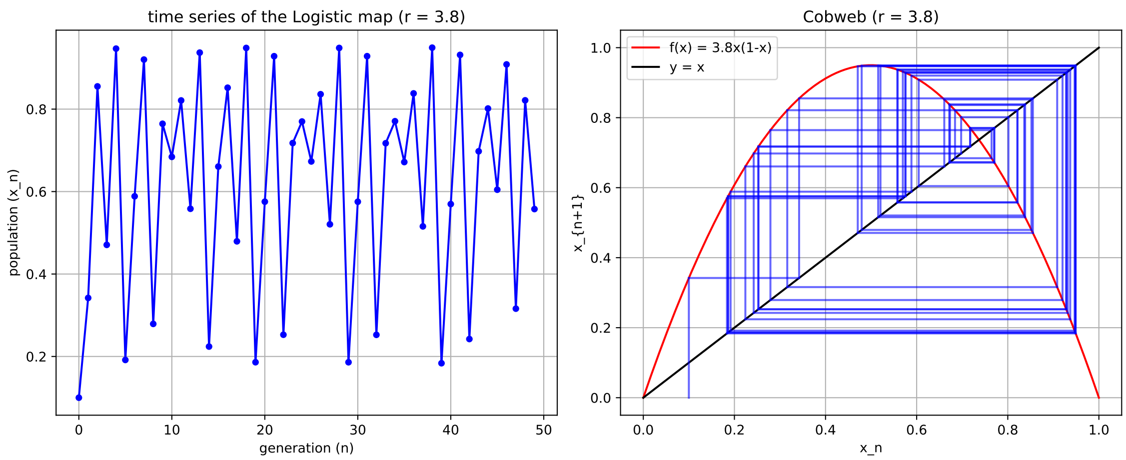

Fig. 3.4. Cobweb diagram for the logistic map with $r = 3.8$.

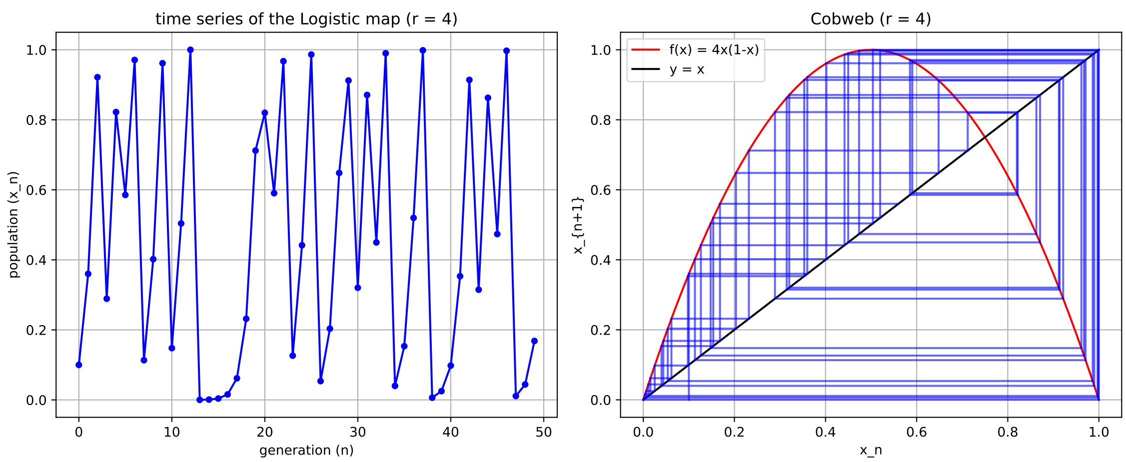

Fig. 3.5. Cobweb diagram for the logistic map with $r = 4$.

Moreover, these results can be visualised clearly in an orbit diagram. The orbit diagram is constructed by evaluating the logistic map over 10,000 equally spaced values of

r in $[3.0, 4.0]$. For each value, the map is iterated 1,000 times from the initial condition $x_0 = 10^{-5}$ to discard the transient and ensure convergence to the asymptotic regime. The following 200 iterates are then plotted as black pixels, so that fixed points appear as isolated curves, periodic orbits as a finite set of branches, and chaotic regimes as dense clouds of points.

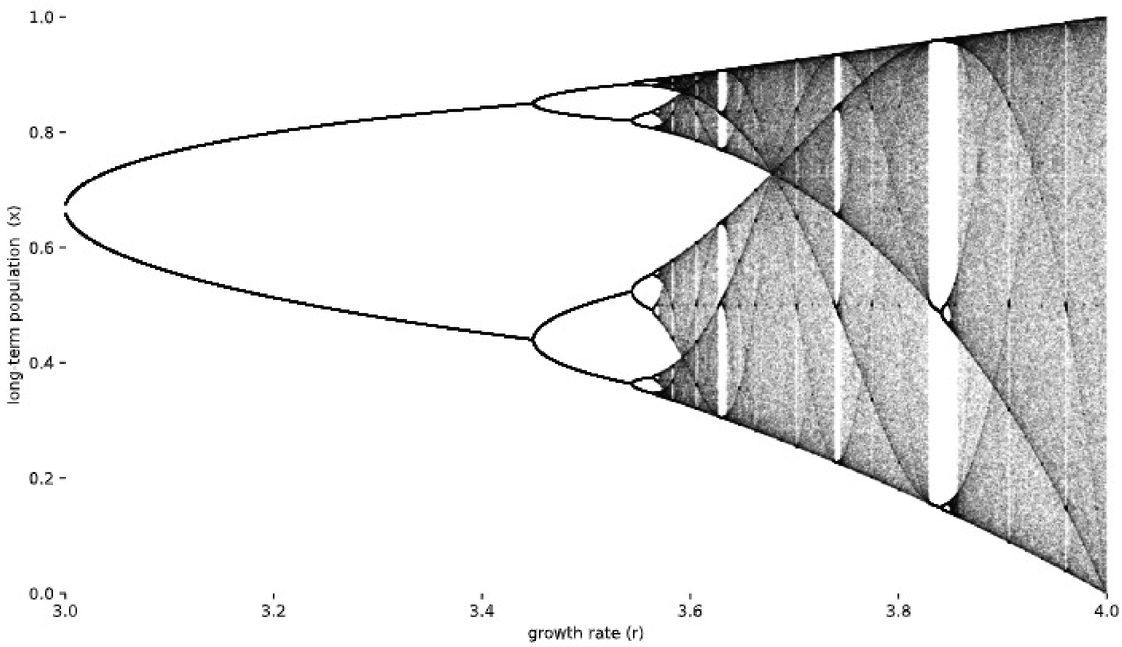

Fig. 3.6. Orbit diagram for the logistic map.

As we can see in the orbit diagram (Figure 3.6), period-doubling happens for the stable values of r of the 2-cycle. After this stable interval, the system enters the chaotic regime. In the diagram, this is represented by the dark shaded areas, which means that the population never settles into a repeating pattern.

However, if we look to the behaviour of the system for further r values, we can see that the dark shades are suddenly interrupted by clear vertical gaps known as *periodic windows*. These are intervals where the system finds order out of chaos, turning back into a stable, predictable cycle; see [21] section 10.4 for further details.

The widest and most famous of these gaps is the period-3 window, which appears near $r \approx 3.8$. Here, the population goes through three distinct values. But this new "stability" is only temporary. As r continues to increase. the systems turns back into chaotic behaviour.

Lyapunov exponent

Let us now analyse how the logistic map is sensitive dependent on the initial conditions for certain values of r.

In order to do so, we need to look at the Lyapunov exponents. From the previous section, we have that the formula for the Lyapunov exponent is:

$$\lambda = \lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x_i)|. \tag{3.53}$$

For the logistic map, the derivative is $f'(x) = r(1-2x)$. Substituting this into the equation yields:

$$\lambda = \lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \ln |r(1-2x_i)|. \tag{3.54}$$

To evaluate this expression numerically, the exponent is computed over 4,000 equally spaced values of r in $[0.1, 4.0]$. For each value, the map is iterated 1,000 times from the initial condition $x_0 = 0.1$ to discard the transient, after which the sum $\sum_{i=0}^{n-1} \ln |r(1-2x_i)|$ is accumulated over $n = 1{,}000$ further iterations. At superstable points, where $f'(x^*) = 0$, the derivative is replaced by the floor value 10^{-12} to avoid evaluating $\ln(0)$, which explains the sharp negative spike visible near $r = 2$ in Figure 3.7. The red dashed line at $\lambda = 0$ separates stable regimes ($\lambda < 0$) from chaotic ones ($\lambda > 0$), and the blue dots mark the specific values of r analysed numerically in the following discussion.

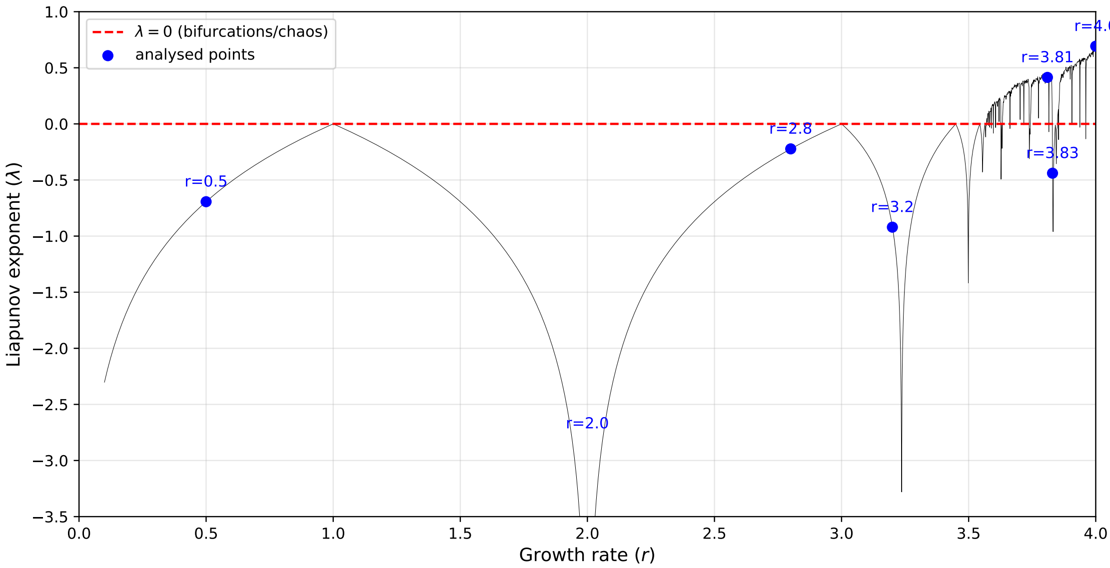

Fig. 3.7. Lyapunov exponent for the logistic map

For the analysed values, we can check numerically these results:

- The system is stable ($\lambda < 0$): $r = 0.50$ ($\lambda = -0.6934$), $r = 2.00$ ($\lambda = -\infty$), $r = 2.80$ ($\lambda = -0.2226$), $r = 3.20$ ($\lambda = -0.9204$), $r = 3.83$ ($\lambda = -0.4395$).
- The system goes under bifurcations ($\lambda = 0$): $r = 1$, $r = 3$, etc. This values are not analysed numerically, but the plot shows that the exponent is exactly 0.
- The system is chaotic ($\lambda > 0$): $r = 3.81$ ($\lambda = 0.4144$), $r = 4.00$ ($\lambda = 0.6925$).

We can observe where the periodic windows appear, since we can observe a difference in the behaviour of the system between values $r = 3.81$ and $r = 3.83$, where the system turns from being chaotic to stable again. This matches with the orbit diagram shown before (Figure 3.6).

We can also clearly check the exact points where bifurcations occur, where the Lyapunov exponent is 0.

Moreover, in the stable regions, not only we have negative exponents, but for certain values, as $r = 2.0$, we have that the exponent is infinitely small. This is interpreted as a *superstable* behaviour (following from [21]).

3.3.2. Universality and Feigenbaum constants

Now that we have analysed how chaotic behaviour happens in a specific example, we can explain how this can be extended to different systems that behave in an identical way thanks to Mitchell J. Feigenbaum [7]. He observed that different recursion relations of the form $x_{n+1} = \lambda f(x_n)$ exhibiting infinite bifurcation possesses a rich quantitative structure that is independent of the specific recursion function.

For this universality to apply, the recursion function must be a *unimodal map* [21]. A unimodal map is a continuous function on an interval that has a single, smooth maximum.

Following Feigenbaum's article [7], this means that, near this unique maximum $\bar{x}$, the function must behave as $f(\bar{x}) - f(x) \sim |x - \bar{x}|^z$ for sufficiently small distances. Thus, the behaviour of the system's dynamics depend on this exponent z.

For the logistic map, this is a parabola, meaning $z = 2$. Therefore, we can check how maps with $z = 2$ behave following the same universal constants.

Example. Observe how the orbit diagram looks for the sine map $x_{n+1} = r\sin(\pi x_n)$. The diagram is constructed following the same methodology as the orbit diagram of the logistic map (Figure 3.6): the map is evaluated over 8,000 equally spaced values of r in $[0.70, 1.00]$, iterated 1,000 times from $x_0 = 10^{-5}$ to discard the transient, and the following 200 iterates are plotted as blue pixels.

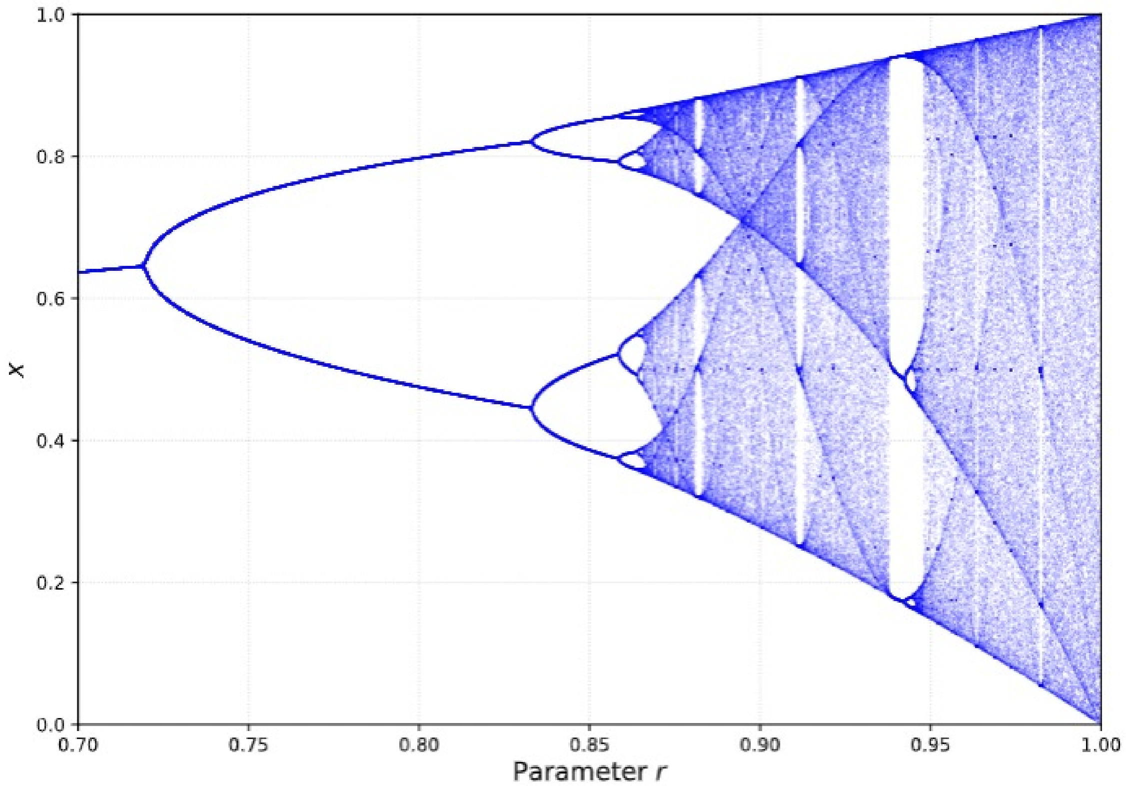

Fig. 3.8. Orbit diagram for the sine map.

To justify why the sine map follows the same universal constants as the logistic map, we must verify that it satisfies the conditions for a unimodal map with a quadratic maximum.

First, we find the critical points of $f(x) = r\sin(\pi x)$ in the interval $[0, 1]$ by setting its first derivative to zero:

$$f'(x) = r\pi\cos(\pi x) = 0. \tag{3.55}$$

In the interval $x \in [0, 1]$, this equation has a unique solution at $\overline{x} = 1/2$. To confirm it is a maximum, we evaluate the second derivative:

$$f''(x) = -r\pi^2\sin(\pi x) \implies f''(1/2) = -r\pi^2 < 0. \tag{3.56}$$

Since $f(x)$ increases on $(0, 1/2)$, decreases on $(1/2, 1)$, and has a unique differentiable maximum at $\overline{x} = 1/2$, it is a *unimodal map*.

Next, we determine the exponent z by analysing the behaviour of the function near its maximum. We apply a Taylor expansion of $f(x)$ around $\overline{x} = 1/2$:

$$f(x) \approx f(\overline{x}) + f'(\overline{x})(x - \overline{x}) + \frac{1}{2}f''(\overline{x})(x - \overline{x})^2. \tag{3.57}$$

Substituting $f(1/2) = r$, $f'(1/2) = 0$, and $f''(1/2) = -r\pi^2$:

$$f(x) \approx r - \frac{r\pi^2}{2}(x - \overline{x})^2. \tag{3.58}$$

Rearranging to match Feigenbaum's condition $f(\overline{x}) - f(x) \sim |x - \overline{x}|^z$:

$$r - f(x) \approx \frac{r\pi^2}{2}|x - 1/2|^2. \tag{3.59}$$

This result confirms that for the sine map, $z = 2$. Because it shares the same quadratic local structure as the logistic map, they follow the same universal behaviour:

There are two fundamental constants that explain these common behaviour among different unimodal maps with same z.

1. **The convergence rate** (δ)

   As seen previously in the orbit diagram (Figure 3.6), the parameter intervals between successive period-doubling bifurcations become progressively smaller. Let b_n be the parameter value at which an element of a stable 2^n-point cycle bifurcates from the original cycle. Feigenbaum discovered that the ratio of the lengths of successive bifurcation intervals converges asymptotically to a universal constant, δ:

   $$\delta = \lim_{n\to\infty} \frac{b_{n+1} - b_n}{b_{n+2} - b_{n+1}}. \tag{3.60}$$

   For any map with $z = 2$, this constant is always $\delta \approx 4.6692016$. This implies that each subsequent period-doubling bifurcation occurs approximately 4.669 times faster than the previous one as the parameter$r$ increases.

2. **The rescaling parameter** (α)

   Moreover, Feigenbaum showed that the local structure of high-order stability sets approaches universality. This happens rescaling in successive bifurcations asymptotically by a constant ratio α. For $z = 2$, this spatial scaling factor is $\alpha \approx 2.5029078$.

This can be checked by observing the pitchfork bifurcations in the orbit diagram. If we measure the distance from the maximum $\overline{x}$ to the nearest element of a stable 2^n-cycle, and compare it to the equivalent distance in the subsequent 2^{n+1}-cycle, the ratio of these distances asymptotically approaches α.

We can visualize these constant values visually from the previous orbit diagram.

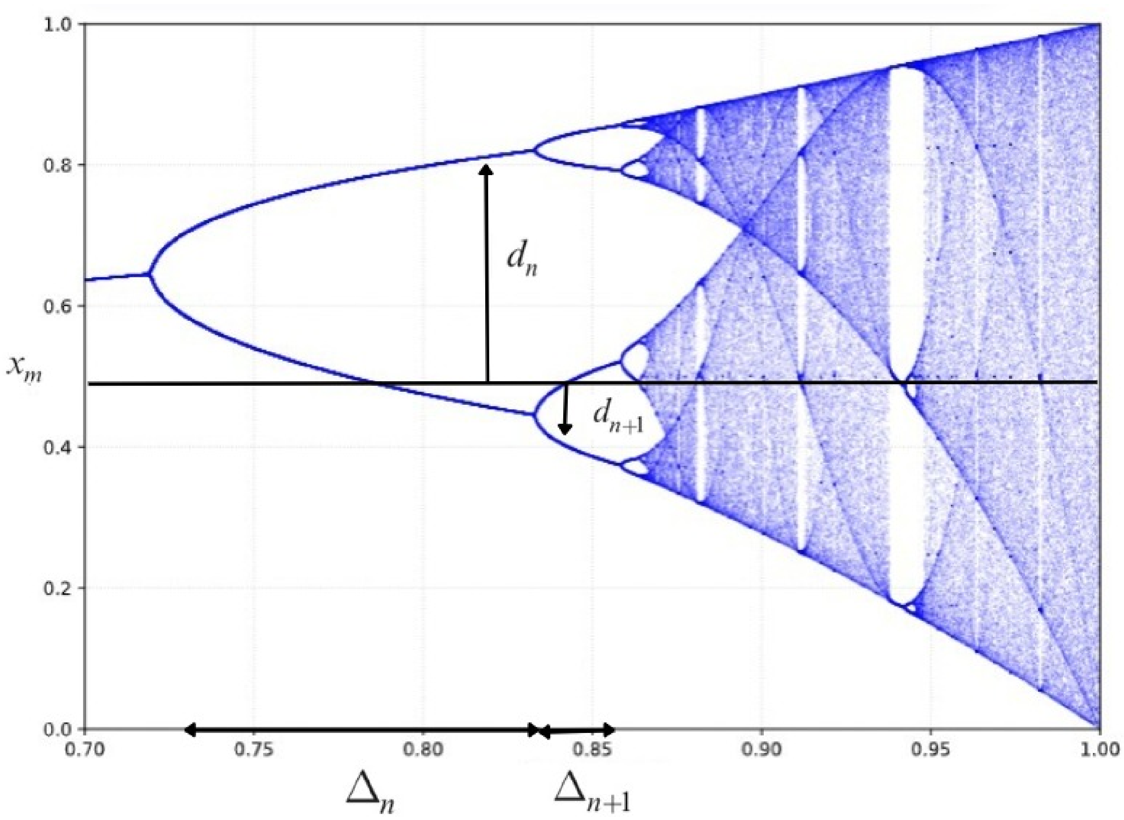

Fig. 3.9. Visualisation of Feigenbaum constants.

This shows how the constants δ and α govern the behaviour of the bifurcation tree. The segment Δ_n represents the width of the parameter window where the 2^n-cycle is stable, such that $\Delta_n = b_{n+1} - b_n$. So that: $\delta = \lim_{n\to\infty} \frac{\Delta_n}{\Delta_{n+1}}$.

On the other hand d_n measures the spatial dimension of the attractors. It represents the vertical distance from the critical point of the map (the maximum x_m) to the nearest branch of the 2^n-cycle. Therefore, we have: $\alpha = \lim_{n\to\infty} \frac{d_n}{d_{n+1}}$.

Renormalization and self-similarity

Now we need to understand the concept of *self-similarity*. A structure is self-similar if zooming in on a small section reveals a geometry identical to the original shape. Feigenbaum demonstrated that the universality of the period-doubling cascade is the direct consequence of this self-similarity, through a recursion mechanism, on the class of functions itself. He adapted the *renormalization group* technique from statistical physics to dynamical systems, allowing us to observe how the dynamics look at different scales.

Let us analyse what happens graphically when a system undergoes a period-doubling bifurcation. Suppose we have a unimodal map $f(x)$ with a maximum at x_m. When the fixed point becomes unstable, a 2-cycle is born. The stability of this new 2-cycle is governed by the second iterate of the map, $f^{(2)}(x) = f(f(x))$.

Feigenbaum observed that, if we look at the central peak of $f^{(2)}(x)$ around the maximum, it is geometrically similar to the original function $f(x)$, but turned upside down and reduced in scale [7] (see Figure 3.10). If we iterate again to find the 4-cycle, the central peak of $f^{(4)}(x) = f^{(2)}(f^{(2)}(x))$ is an inverted, smaller version of the peak of $f^{(2)}(x)$.

At each bifurcation n, we can draw a small box around the central peak of $f^{(2^n)}(x)$. If we magnify this tiny box and invert it, it perfectly overlays the larger box from the previous generation $f^{(2^{n-1})}(x)$ [21]. In all three figures below, each left panel shows the original and iterated functions together with a green dashed box enclosing the central peak to be renormalised, and each right panel shows the result of applying the renormalisation transformation: the contents of the box are rescaled horizontally to fill the unit interval and inverted vertically, so that the resulting curve (green dashed line) overlays the original function. Both functions are evaluated on a uniform grid of 1,000 points.

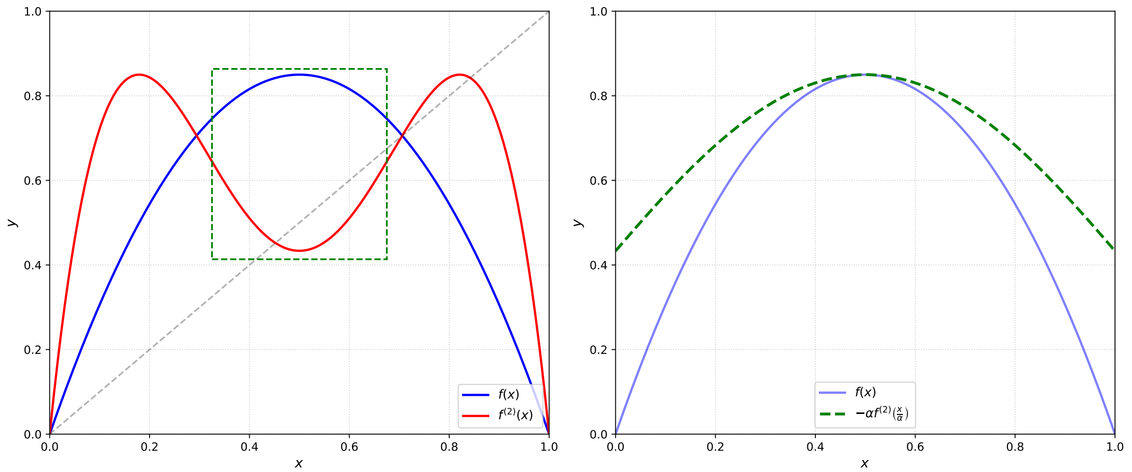

Fig. 3.10. Renormalization of the logistic map with $r = 3.4$.

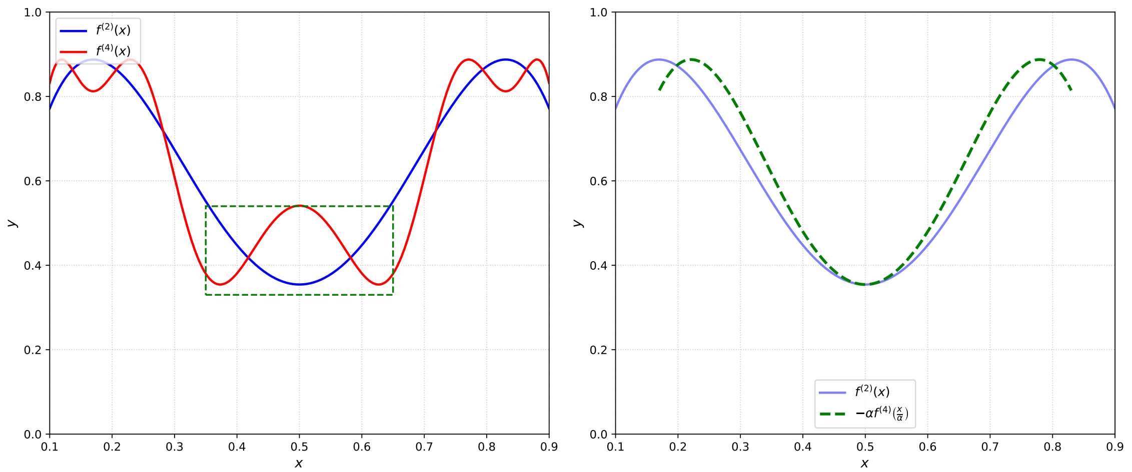

Fig. 3.11. Renormalization of the second iterate of the logistic map with $r = 3.4$.

To express this process, we can relate the original map evaluated at a superstable parameter R_0 to its second iterate evaluated at the next superstable parameter R_1.

Remark. A period-k cycle is said to be *superstable* if one of its points lands exactly on the maximum of the map (the critical point x_m). That is why the perfect alignment is achieved with this parameter values.

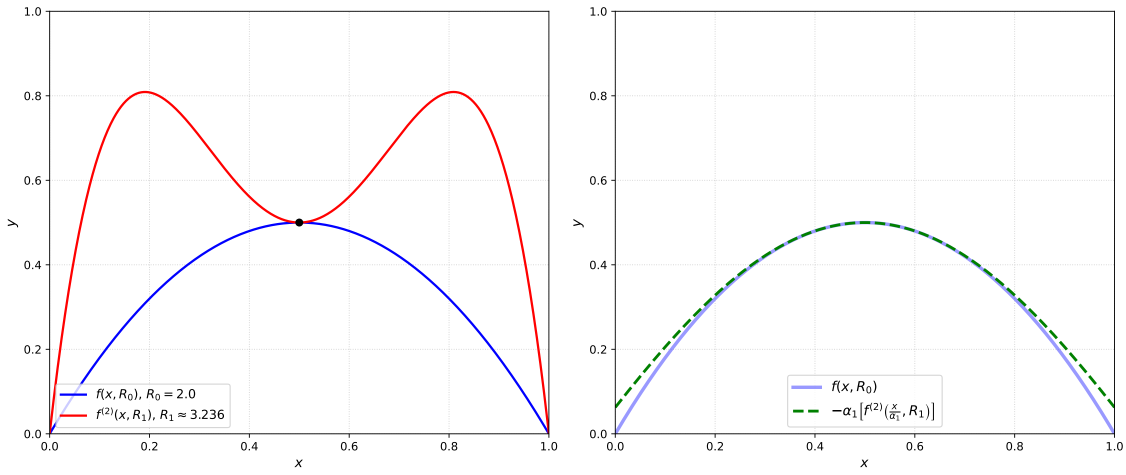

Fig. 3.12. Renormalization of the logistic map with R_0 and R_1.

Now, we can check the analytical process; see [21]. First, we scale the spatial input by α and the amplitude by $-\alpha$:

$$f(x, R_0) \approx -\alpha f^{(2)}\left(\frac{x}{\alpha}, R_1\right). \tag{3.61}$$

We can apply the exact same logic to the fourth iterate at the subsequent superstable parameter R_2:

$$f^{(2)}\left(\frac{x}{\alpha}, R_1\right) \approx -\alpha f^{(4)}\left(\frac{x}{\alpha^2}, R_2\right). \tag{3.62}$$

Substituting this second relation back into the first equation yields:

$$f(x, R_0) \approx (-\alpha)^2 f^{(4)}\left(\frac{x}{\alpha^2}, R_2\right). \tag{3.63}$$

After repeating this renormalization procedure n times, we arrive at the general relation:

$$f(x, R_0) \approx (-\alpha)^n f^{(2^n)}\left(\frac{x}{\alpha^n}, R_n\right). \tag{3.64}$$

Feigenbaum found numerically that as $n \to \infty$, this sequence of functions converges to a limit function:

$$\lim_{n\to\infty} (-\alpha)^n f^{(2^n)}\left(\frac{x}{\alpha^n}, R_n\right) = g^*(x), \tag{3.65}$$

where $g^*(x)$ is a *universal function*.

Remark. This limit function only exists for the correct α universal value.

At the accumulation point of the cascade (R_∞), the parameter r no longer needs to be shifted. The limiting universal function $g^*(x)$ acts as a *fixed point* in function space under this operator, meaning it maps onto itself:

$$g^*(x) = -\alpha g^*\left(g^*\left(\frac{x}{\alpha}\right)\right). \tag{3.66}$$

The function must satisfy appropriate boundary conditions:

1. By shifting the origin, we place the maximum of our maps at $x = 0$, which implies $g^{*'}(0) = 0$.
2. We also normalize the height of the peak to 1 without loss of generality, setting $g^*(0) = 1$.

Evaluating the functional equation at $x = 0$ yields:

$$g^*(0) = -\alpha g^*(g^*(0)) \implies 1 = -\alpha g^*(1) \implies \alpha = -\frac{1}{g^*(1)}. \tag{3.67}$$

This result shows that the universal scale factor α is completely determined by the function $g^*(x)$ itself. Feigenbaum solved it using a power series expansion, $g^*(x) \approx 1 + c_2x^2 + c_4x^4 + \dots$. By substituting this series into the functional equation (3.66) and matching the coefficients, he obtained the universal values $\alpha \approx 2.5029$ and $g^*(x) \approx 1 - 1.527x^2 + 0.104x^4 + \dots$.

Consequently, the shape of any unimodal map with a quadratic maximum ($z = 2$) is attracted toward a high-dimensional surface that leads directly to this specific $g^*(x)$.

This yields to a wonderful universal result: at the edge of chaos, all such maps have lost their individual global characteristics and have turned into the exact same universal geometry.

3.4. Fractal geometry

As we have seen in the previous section, one characteristic of chaotic systems is *self-similarity* (zooming into the bifurcation shows the same features as the whole). This repetition of structure across different scales reveals a completely new geometry.

For centuries, science described the natural world using Euclidean geometry: lines, planes, circles, and spheres. However, Benoit Mandelbrot questioned this fact: "Clouds are not spheres, mountains are not cones, coastlines are not circles, and bark is not smooth, nor does lightning travel in a straight line" [15].

In 1975, Mandelbrot proposed the term *fractal* (from the Latin *fractus*, meaning fractured or broken) to describe a new geometry of nature. A fractal is a *fragmented* geometric shape that can be split into parts, each of which is (at least approximately) a copy at a lower scale of the whole. Fractals show a complex structure at every level of magnification. In the context of non-linear dynamics, this geometry is the fundamental pattern that chaotic systems follow in the phase space.

3.4.1. Fractal dimension

This new geometry yielded some paradoxes, especially in the context of *dimensions*. In Euclidean geometry, dimension is an integer: a point is 0-dimensional, a line is 1-dimensional, a plane is 2-dimensional, and a solid is 3-dimensional. However, fractal structures do not follow this rules, possessing a fractional dimension [15].

Example. To illustrate this fact, we can analyse the von Koch curve.

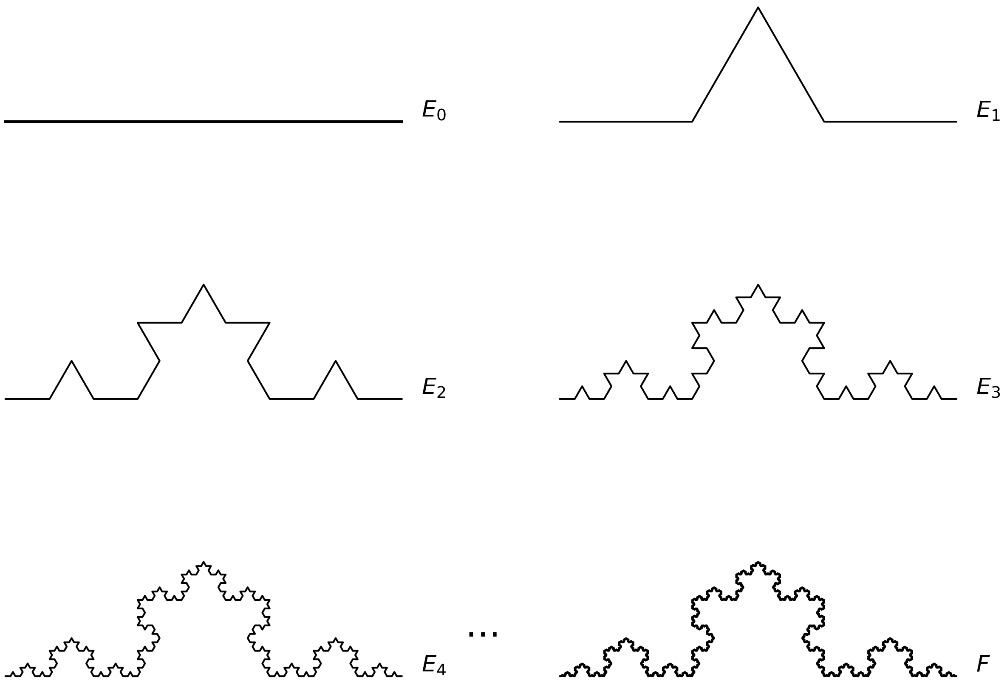

Fig. 3.13. Construction of the von Koch curve. At each stage, the middle third of each interval is replaced by the other two sides of an equilateral triangle.

Regarding Euclidean geometry, a standard line is one-dimensional. The von Koch curve, however, does not fit in this idea: it is a continuous curve of infinite length confined within a finite area, with zero area itself. Consequently, it occupies more space than a standard one-dimensional line, but not enough to be considered a two-dimensional surface.

To explain this phenomenon, we must define the concept of **fractal dimension**.

Remark. The *Cantor set* is also a great example of this topic and topologically quite interesting, however, for simplicity it is not going to be analysed in this manuscript.

The box-counting dimension

The most widely used metric in numerical experiments and dynamical systems is the *box-counting dimension* [21].

Suppose we have a complex geometric set S in a Euclidean space. We cover this set with a uniform grid of boxes of side length ϵ. Let $N(\epsilon)$ be the minimum number of boxes required to cover the set completely. For standard geometric objects, the number of boxes scales according to a simple power law: $N(\epsilon) \approx \epsilon^{-D}$, where D is the integer dimension of the object. For instance, a simple line segment of length L requires $N(\epsilon) \approx L/\epsilon$ boxes ($D = 1$), while a flat area A requires $N(\epsilon) \approx A/\epsilon^2$ boxes ($D = 2$). We can see figure 3.14 for a visualization of the idea.

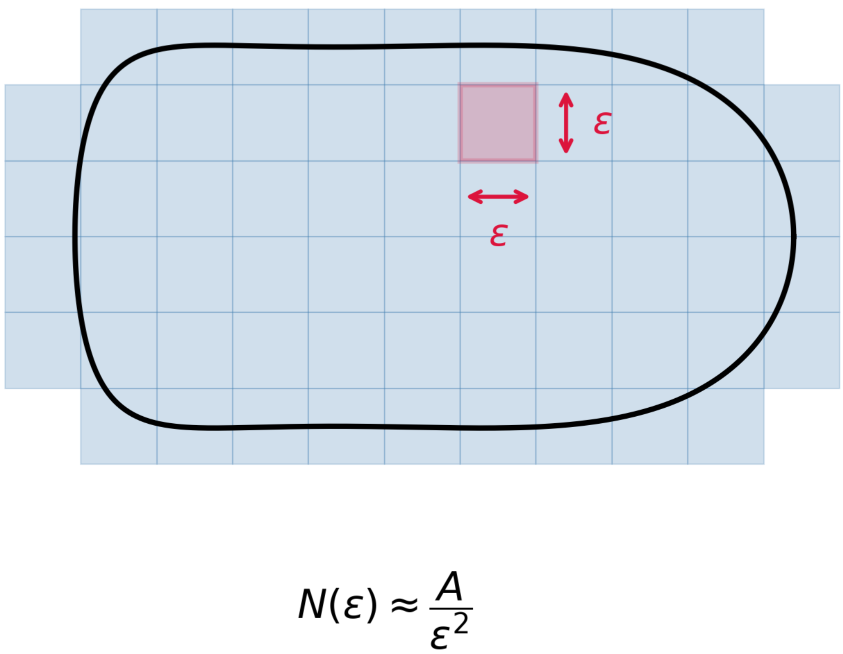

Fig. 3.14. Box-counting technique.

By taking the natural logarithm on both sides of the relationship $N(\epsilon) \approx c\epsilon^{-D}$ (where c is a constant), we can isolate D. Then we have:

$$\ln(N(\epsilon)) = \ln(c) - D\ln(\epsilon).$$

Rearranging the terms to solve for D, we obtain:

$$D = \frac{\ln(N(\epsilon))}{-\ln(\epsilon)} - \frac{\ln(c)}{-\ln(\epsilon)}.$$

Using the logarithmic property $-\ln(\epsilon) = \ln(1/\epsilon)$, this expression can be rewritten. Moreover, as $\epsilon \to 0$, the denominator $\ln(\epsilon)$ approaches $-\infty$. Consequently, the term containing the constant c vanishes $\left(\frac{\ln(c)}{-\infty} \to 0\right)$. The box-counting dimension is defined as the remaining limit:

$$D = \lim_{\epsilon \to 0} \frac{\ln N(\epsilon)}{\ln(1/\epsilon)}. \tag{3.68}$$

Now we can go back to our example to compute its box-counting dimension.

As illustrated in Figure 3.13, at each iteration step, the length of the basic segments is reduced by a factor of 3. Therefore, our measuring scale is $\epsilon = 1/3$. At the same time, each original straight segment is replaced by 4 new, smaller segments. Thus, the number of pieces required to cover the new structure increases by a factor of 4, yielding $N = 4$.

Substituting these values into the dimension formula, we get:

$$D = \frac{\ln(4)}{\ln(3)} \approx 1.2619. \tag{3.69}$$

Correlation dimension

Calculating the box-counting dimension of a strange attractor requires an exponentially growing number of boxes as the grid size $\epsilon \to 0$. In higher dimensions, most of these boxes will be completely empty, making the algorithm highly inefficient and memory-intensive [21].

Peter Grassberger and Itamar Procaccia introduced a more computationally efficient metric in 1983 known as the *correlation dimension* (D_c) [9].

Suppose we have an experimental time series or a numerical simulation consisting of N points, $\{\vec{x}_1, \vec{x}_2, \ldots, \vec{x}_N\}$, lying on an attractor in the phase space. The correlation dimension measures the probability that two randomly chosen points are separated by a distance less than a given radius r. To compute this, we define the *correlation sum* $C(r)$ as:

$$C(r) = \lim_{N \to \infty} \frac{2}{N(N-1)} \sum_{i=1}^{N} \sum_{j=i+1}^{N} H(r - ||\vec{x}_i - \vec{x}_j||) \tag{3.70}$$

Here, H is the Heaviside step function ($H(x) = 1$ if $x \geq 0$, and 0 otherwise), and $||\vec{x}_i - \vec{x}_j||$ represents the Euclidean distance between the pair of points. The term $\frac{2}{N(N-1)}$ is simply a normalization factor representing the total number of distinct point pairs. Essentially, $C(r)$ counts the fraction of point pairs whose distance is less than r.

For fractal attractors, as the radius r becomes very small ($r \to 0$), the correlation sum scales according to a power law:

$$C(r) \approx r^{D_c} \tag{3.71}$$

By taking the natural logarithm of both sides, we arrive at the definition of the correlation dimension:

$$D_c = \lim_{r \to 0} \frac{\ln C(r)}{\ln r} \tag{3.72}$$

In practical applications, D_c is determined geometrically. By plotting $\ln C(r)$ against $\ln r$ for various values of r, one can identify a linear scaling region. The slope of this line directly yields the correlation dimension D_c [6].

Example. We can evaluate this algorithm on the logistic map ($x_{n+1} = rx_n(1 - x_n)$). We analyse the system at the accumulation point of period-doubling bifurcations,($r_\infty \approx 3.5699456$).

At this critical parameter, the system is at the exact threshold where chaos begins. The resulting invariant set is not a continuous line, nor a finite set of periodic points, but an infinitely disconnected fractal dust. The attractor is generated by iterating the map for $N = 10{,}000$ steps after discarding a transient of 10,000 iterations from the initial condition $x_0 = 0.5$. The correlation sum is then computed using the pairs (x_n, x_{n+1}), evaluated over 40 logarithmically spaced values of r between r_{min} and $r_{\text{max}}/3$. The correlation dimension D_c is estimated as the slope of the linear scaling region, identified in the interval $\ln r \in [-4.5, -2.0]$.

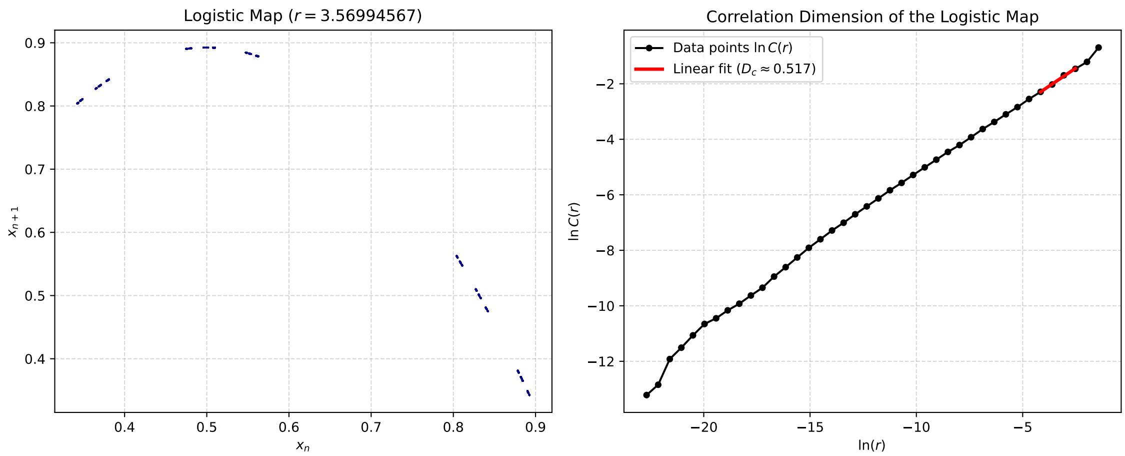

Fig. 3.15. Correlation dimension analysis of the logistic map at $r \approx 3.5699456$

As shown in Figure 3.15, our numerical application of the correlation sum yields a dimension of $D_c \approx 0.50$.

More interesting is the analysis of continuous differential equations with fractal geometry characteristics.

Example. We have numerically simulated a sample of points from the Lorenz system (with $\sigma = 10$, $\rho = 28$, and $\beta = 8/3$) and computed its correlation dimension using the Grassberger–Procaccia algorithm. The attractor is sampled by integrating the system with a fourth-order Runge–Kutta scheme with internal step size $\Delta t = 0.01$, discarding an initial transient of 5,000 steps to ensure convergence to the attractor. A total of $N = 15{,}000$ points are then recorded at a sampling interval of $\tau = 0.25$ time units (corresponding to 25 internal steps per saved point), chosen to reduce temporal correlations between consecutive samples. The correlation sum $C(r)$ is evaluated over 40 logarithmically spaced values of r between $r_{\min}$ and $r_{\max}/3$, and the correlation dimension D_c is estimated as the slope of the linear region of the $\ln C(r)$ vs $\ln r$ plot. Figure 3.16 displays both the sampled points revealing the shape of the strange attractor in the (x, z) plane and the resulting log-log plot.

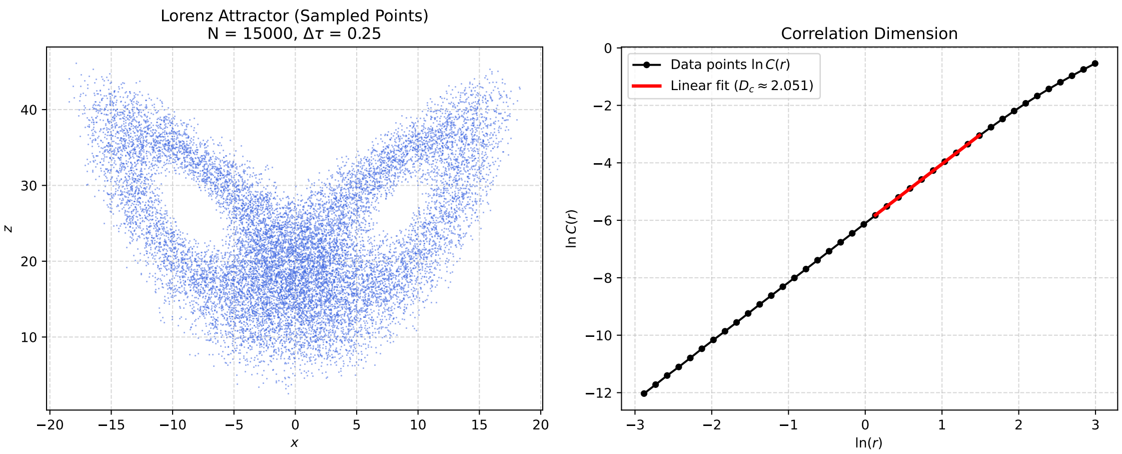

Fig. 3.16. Grassberger–Procaccia algorithm for the Lorenz system.

These results yield a correlation dimension of $D_C \approx 2.051$, consistent with the value of 2.05 ± 0.01 established by Grassberger and Procaccia under identical standard parameters ($\sigma = 10, \rho = 28, \beta = 8/3$).

A correlation dimension between 2 and 3 proves that the attractor is not a simple surface, but a complex structure. This fact leads us to the formal definition of one of the most fascinating objects in non-linear dynamics: the Strange Attractor.

3.4.2. Strange attractors

In order to define the strange attractor, we will focus on the properties of the Lorenz system (previously analysed in Section 3.1). We concluded that the system was strictly volume-dissipative. The divergence of the vector field $V(x, y, z)$ is everywhere negative and constant:

$$\nabla \cdot V = \frac{\partial \dot{x}}{\partial x} + \frac{\partial \dot{y}}{\partial y} + \frac{\partial \dot{z}}{\partial z} = -(\sigma + 1 + \beta) < 0 \tag{3.73}$$

By Liouville's theorem, this implies that any initial volume V_0 in the phase space contracts exponentially over time as $V(t) = V_0 e^{-(\sigma+1+\beta)t}$. As $t \to \infty$, the volume of the phase flow shrinks to zero.

A system whose volume contracts to zero typically settles into a zero-dimensional point (a stable equilibrium) or a one-dimensional closed loop (a limit cycle). However, for the chaotic parameter regime, the Lorenz system does neither. The trajectories never intersect, nor do they settle into a periodic orbit. Instead, they are asymptotically attracted to a bounded region of zero volume but infinite topological complexity: the **Strange Attractor**.

This is the effect of the mechanism of *stretching and folding*. Driven by a positive largest Lyapunov exponent, nearby trajectories are exponentially stretched apart. Simultaneously, the volume dissipation forces the phase space to fold over itself to remain bounded.

Visually, this means that the phase space is stretched out, folded over, and compressed infinitely many times. The result is a fractal structure with a fractal dimension (approximately $D_H \approx 2.06$ for the standard parameters, as we discussed in the previous section).

Figure 3.17 illustrates the global geometry of this attractor, famously known as the Lorenz butterfly. The trajectory is obtained by integrating the system with the adaptive RK45 method over $t \in [0, 100]$, evaluated at 100,000 equally spaced time points from the initial condition $(x_0, y_0, z_0) = (1, 1, 1)$. The first 2,000 points are discarded as transient, and the remaining trajectory is plotted as a single continuous curve.

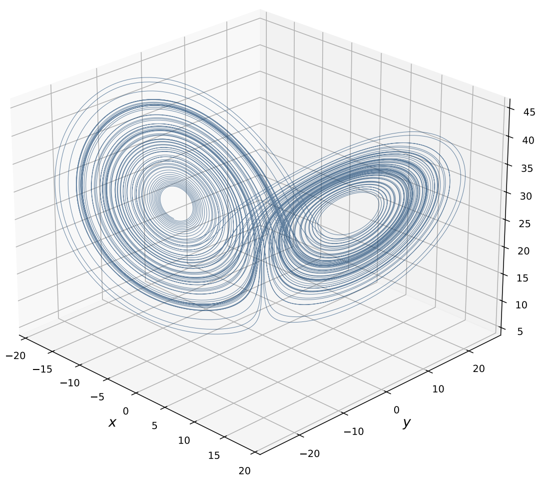

Fig. 3.17. Global view of the Lorenz strange attractor ($\sigma = 10, \rho = 28, \beta = 8/3$).

4. SIMULATING CHAOTIC BEHAVIOUR: APPLICATION ON RUMOUR SPREADING

4.1. Modelling the spread of rumours on social media

4.1.1. Background and motivation

To understand how rumours spread, researchers have taken advantage of tools from the study of epidemics. The comparison makes sense, since information spreading through a network of people who have not heard it yet acts like a virus infecting a healthy population (see [17]).

Classic frameworks, like the Daley-Kendall SIR model, explain this process by grouping people into three simple categories: *ignorants* (who have not heard the news), *spreaders* (who are actively sharing it), and *stiflers* (who know about it but have stopped caring or sharing). By using mathematical equations, researchers can track how people move between these groups as the rumour runs its course [22].

These models have produced useful insights into the dynamics of rumour spread, but they share an assumption that is difficult to justify in the context of modern social media. Both models treat the transmission rate (the rate at which spreaders come into effective contact with ignorants) as a constant. In a face-to-face social network this is a reasonable approximation, since human contact rates vary slowly and are determined by social relationships. However, on a digital platform, the effective rate at which content reaches users is not determined only by social proximity. It is also modulated by the platform's recommendation algorithm instead, which decides at each moment which content to show on each user's feed. This introduces a temporal variability into the transmission rate that the classical models do not capture.

Researchers have actually observed these changes in the real world. Huszár et al. [10] showed that recommendation algorithms amplify content in ways that depend on engagement metrics and that vary over time. Wu and Huberman [23] established that collective attention on digital platforms decays rapidly as novelty is exhausted, and that new content periodically displaces old content from users' attention. Asur et al. [1] documented that trending topics on social media exhibit strong competition and short persistence, with periodic resurgences driven by external events and algorithmic promotion cycles. Taken together, these findings suggest that the effective transmission rate of a rumour on a social platform oscillates over time, rising when the algorithm amplifies the content and falling when it is displaced by newer material.

A second limitation of the classical models is the transition from ignorance to active spreading. In the SIR model, a user who encounters the rumour immediately becomes an active spreader. Nevertheless, empirical studies of online behaviour show that a large fraction of platform users consume content without producing it, as they can read and react to posts without sharing them (see [17]). These users represent an important part of the population that the classical models ignore.

The model proposed in this section addresses both limitations. It is a modified SEIR model in which the transmission rate is a periodic function of time and the population includes an intermediate state between susceptibles and infected, which is the exposed population. The result is a non-autonomous system whose dynamics can offer powerful insights on how rumour spread happens in digital platforms.

4.1.2. Starting point: the classical SIR model

The natural starting point for the derivation is the classical SIR model (see [3]), which partitions a closed population of size N into susceptible individuals $S(t)$, who have not yet encountered the rumour; infected individuals $I(t)$, who know the rumour and are actively spreading it; and recovered individuals $R(t)$, who have lost interest and no longer participate in the diffusion. With $S + I + R = N$, the dynamics are governed by

$$\begin{cases} \dfrac{dS}{dt} = -\beta S I, \\ \dfrac{dI}{dt} = \beta S I - \gamma I, \\ \dfrac{dR}{dt} = \gamma I. \end{cases} \tag{4.1}$$

where $\beta > 0$ is the transmission rate and $\gamma > 0$ is the recovery rate. In the rumour context, β measures the rate at which an active spreader transmits the rumour to a susceptible user per unit time, and γ is the rate at which spreaders spontaneously stop sharing the content.

This model has two structural limitations for social media platforms. The first is that R is an absorbing state, since once a user stops spreading the rumour they never return to the susceptible state, and the rumour eventually dies out permanently. On social media, however, users who have previously seen content can be exposed to it again when the algorithm resurfaces it, and their attention to the topic can be renewed. The second limitation is that β is constant, which, as argued above, fails to capture the periodic amplification behaviour of recommendation algorithms.

4.1.3. The proposed model

We propose a non-autonomous SEIR model that addresses both limitations. This model is based on the mathematical framework analysed in [4], we assume a normalised and demographically open population in which $S + E + I + R = 1$ at all times, so that each variable represents a fraction of the total user base. The recovered compartment R is determined implicitly by $R = 1 - S - E - I$, so the system is described by three equations.

The four compartments have the following interpretations in the social media context.

- $S(t)$ represents the fraction of users who have not yet been exposed to the rumour. This is not a fixed quantity, since new susceptible users continuously enter the platform.
- $E(t)$ represents the fraction of users who have encountered the rumour but have not yet decided to share it (called *lurkers*). The biological comparison of this state would be the incubation period, during which an individual is infected but not yet infectious.
- $I(t)$ represents the fraction of users who are actively spreading the rumour. Both positive and negative engagement, sharing to agree and sharing to refute, contribute to this compartment, since both increase the content's visibility.
- $R(t)$ represents the fraction of users who have lost interest in the rumour. They no longer produce engagement that would amplify the rumour, though they may still see it occasionally if the algorithm resurfaces it.

The five parameters of the model have the following interpretations.

- μ is the demographic turnover rate of the platform. Its inverse $1/\mu$ is the mean time a user spends active on the platform before becoming inactive.
- β_0 is the baseline transmission rate in the absence of algorithmic forcing. It represents the rate at which an active spreader causes a susceptible user to become exposed to the rumour under average conditions, combining the effects of follower network structure, organic reach, and baseline algorithmic amplification. A large β_0 corresponds to highly viral content with a wide organic reach; a small β_0 corresponds to content that spreads slowly even without algorithmic suppression.
- σ is the rate at which lurkers become active spreaders. Its inverse $1/\sigma$ is the mean time a user spends in the exposed compartment before deciding to share the content. A large σ corresponds to impulsive sharing behaviour with a short deliberation time; a small σ corresponds to content that users observe for a long time before acting, such as complex information that requires verification.

- γ is the disengagement rate, the rate at which active spreaders stop sharing the rumour. Its inverse $1/\gamma$ is the mean duration of active engagement for a spreader. On social media, where attention cycles are short and topics are rapidly displaced by new content, γ is expected to be large: a value of $\gamma = 25\ \mathrm{yr}^{-1}$ corresponds to a mean engagement duration of approximately two weeks, which is consistent with the empirical observation that most social media trends peak and decline within days to weeks (see [1]).

- ε is the forcing amplitude, the most important parameter for the purposes of this chapter. It is defined by:

$$\beta(t) = \beta_0 \left(1 - \varepsilon \cos(2\pi t)\right), \tag{4.2}$$

  where time is measured in years. The transmission rate oscillates between a maximum of $\beta_0(1+\varepsilon)$, reached when the algorithm is most actively promoting the content, and a minimum of $\beta_0(1-\varepsilon)$, reached at the opposite phase when the content receives minimal algorithmic support. The amplitude ε therefore measures the relative intensity of algorithmic intervention. In other words, how much the platform's promotion cycles amplify or suppress the rumour's reach compared to its organic baseline. The period of one year reflects the annual cycle of social media activity, driven by advertising calendars, political events, cultural seasons, and institutional rhythms that are documented in [1]. A higher frequency such as weekly or daily cycles are not included in the model for simplicity.

The dynamics of the system are governed by:

$$\begin{cases} \dfrac{dS}{dt} = \mu(1-S) - \beta(t)SI, \\ \dfrac{dE}{dt} = \beta(t)SI - (\mu+\sigma)E, \\ \dfrac{dI}{dt} = \sigma E - (\mu+\gamma)I. \end{cases} \tag{4.3}$$

These equations describe the flow of users between compartments through specific transition rates:

- The term $\mu(1-S)$ represents the net influx of new users joining the platform. The term $\beta(t)SI$ is the transmission rate, where susceptible users encounter the rumour and move to the exposed compartment.

- Exposed users arrive at the rate $\beta(t)SI$ and leave at the rate $(\mu+\sigma)E$, either by leaving the platform (μ) or by transitioning into active spreaders (σ).

- The spreader population grows from the newly active users (σE) and decreases at a rate $(\mu+\gamma)I$ as spreaders either leave the platform or lose interest in the rumour.

4.2. Bifurcation analysis under algorithmic forcing

The bifurcation diagram is the main tool used in this chapter to analyse the long-run behaviour of the forced SEIR system as the algorithmic forcing amplitude ε varies. The diagram records only the local maxima of $I(t)$ in the stationary regime, plotting each peak against the value of ε that produced it. A single isolated point at a given ε indicates a periodic orbit with one dominant peak per cycle, while a pair of points indicates period-doubling. Finally, a set of points with no patterns could indicate chaotic dynamics.

4.2.1. Parameter selection and case definitions

Four parameter configurations are studied. Each configuration fixes a set of values for β_0, γ, σ and μ, which represent different engagement dynamics (different types of social media platforms or users). The transmission rate is fixed at $\beta_0 = 1241\,\text{yr}^{-1}$ across all four cases, so that the only source of variation is the structure of user behaviour (γ, σ and μ). The four cases are summarised in Table 4.1.

Case	$\gamma\,(\text{yr}^{-1})$	σ	$\mu\,(\text{yr}^{-1})$	Mean engagement
Case 1	120	365/3	0.10	≈ 3 days
Case 2	73	365/8	0.02	≈ 5 days
Case 3	36	365/15	0.01	≈ 10 days
Case 4	18	365/25	0.005	≈ 20 days

TABLE 4.1. PARAMETER CONFIGURATIONS FOR THE FOUR CASES STUDIED.

Case 1 represents the fastest engagement dynamics of the four configurations. With $\gamma = 120\,\text{yr}^{-1}$ and $\sigma = 365/3$, both the latency period and the mean spreading duration are approximately three days. The user renewal rate is also the highest of the four cases ($\mu = 0.10$), showing high demographic turnover where users cycle through content rapidly and move on quickly.

Case 2 uses intermediate engagement parameters, with $\gamma = 73\,\text{yr}^{-1}$ (mean engagement ≈ 5 days), $\sigma = 365/8$ (latency ≈ 8 days), and $\mu = 0.02$. These are the reference parameters studied in [4], representing a general environment with moderate engagement dynamics and a stable but actively renewing user base.

Case 3 has a slower engagement timescale , with $\gamma = 36\,\text{yr}^{-1}$ (mean engagement ≈ 10 days), $\sigma = 365/15$ (latency ≈ 15 days), and $\mu = 0.01$. Active spreaders remain in the infectious compartment approximately twice as long as in Case 2 before disengaging, and the user base turns over more slowly.

Case 4 represents the slowest and most persistent engagement environment considered.

With $\gamma = 18\,\text{yr}^{-1}$ (mean engagement $\approx$ 20 days), $\sigma = 365/25$ (latency $\approx$ 25 days), and $\mu = 0.005$, this case has the smallest demographic turnover and the longest individual spreading events of the four cases.

4.2.2. Temporal setup and computational methodology

All simulations share the same time structure. The system is integrated numerically using `scipy.integrate.odeint` with absolute and relative tolerances of 10^{-10} and 10^{-8} respectively, and a maximum step size of $\Delta t = 0.02\,\text{yr}$, ensuring adequate resolution of the fast components of the forcing.

A transient of $t_{\text{trans}} = 950$ years is discarded at the start of each simulation. This interval is chosen to guarantee that the solution has converged to its asymptotic attractor, regardless of the value of ε. The analysis is then performed over a sampling window of $t_{\text{samp}} = 50$ years, giving a total integration time of 1000 years per run.

The bifurcation diagram is constructed by varying ε over 250 equally spaced values in $[0, 0.30]$. For each value, the local maxima of $I(t)$ in the sampling window are identified using `scipy.signal.find_peaks`, and each peak value I^* is recorded as $\log_{10}(I^*)$. The initial condition is fixed at $S(0) = 0.06$, $E(0) = 0.001$, $I(0) = 0.001$ for all runs. The same transient and sampling window are used consistently across the bifurcation diagrams and the phase portraits of the following section, ensuring that the two analysis are directly comparable.

4.2.3. Analysis of the bifurcation diagrams

Case 1

Figure 4.1 shows the bifurcation diagram for Case 1.

For low values of ε, the system produces a single uniform peak of active spreaders per cycle. As the forcing amplitude increases, the system alternates between two different peak heights, appearing as a splitting into two distinct branches around $\varepsilon \approx 0.05$. These branches merge back near $\varepsilon \approx 0.20$.

For $\varepsilon > 0.25$, the dynamics change entirely, since a new cluster of low peaks emerges near $\log_{10}(I^*) \approx -9$. These represent near-extinction events where the rumour almost vanishes before the next algorithmic amplification cycle revives it.

This behaviour is driven by the high disengagement rate ($\gamma = 120$). Because spreading waves burn out in approximately three days, the algorithmic forcing cycle lacks enough time to interact with the system and build up more complex dynamics.

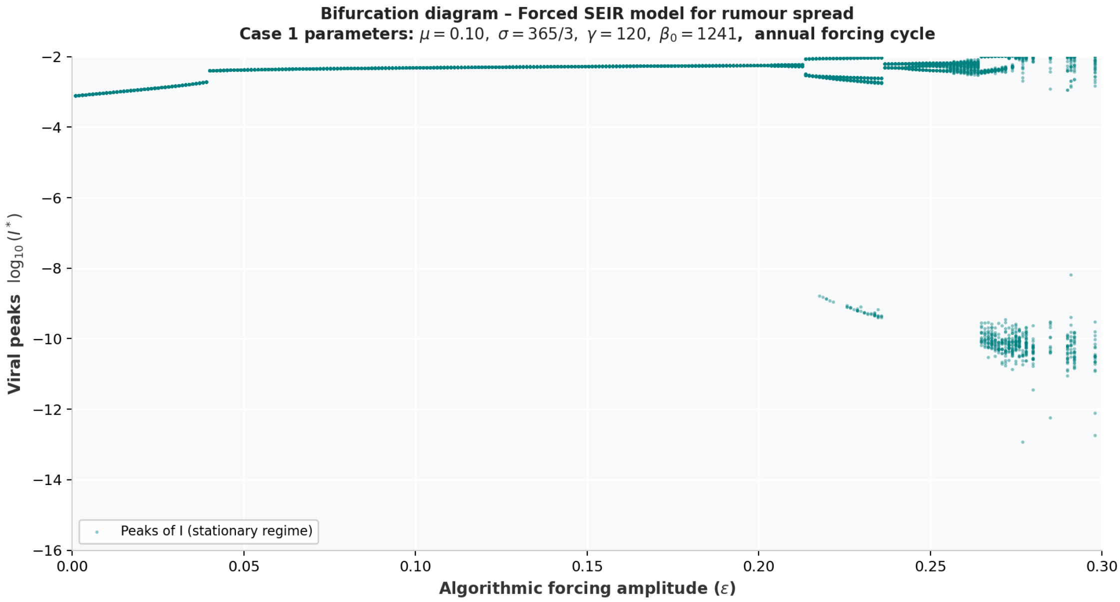

Fig. 4.1. Bifurcation diagram for Case 1 ($\gamma = 120, \sigma = 365/3, \mu = 0.10, \beta_0 = 1241$).

Case 2

Figure 4.2 shows the bifurcation diagram for Case 2.

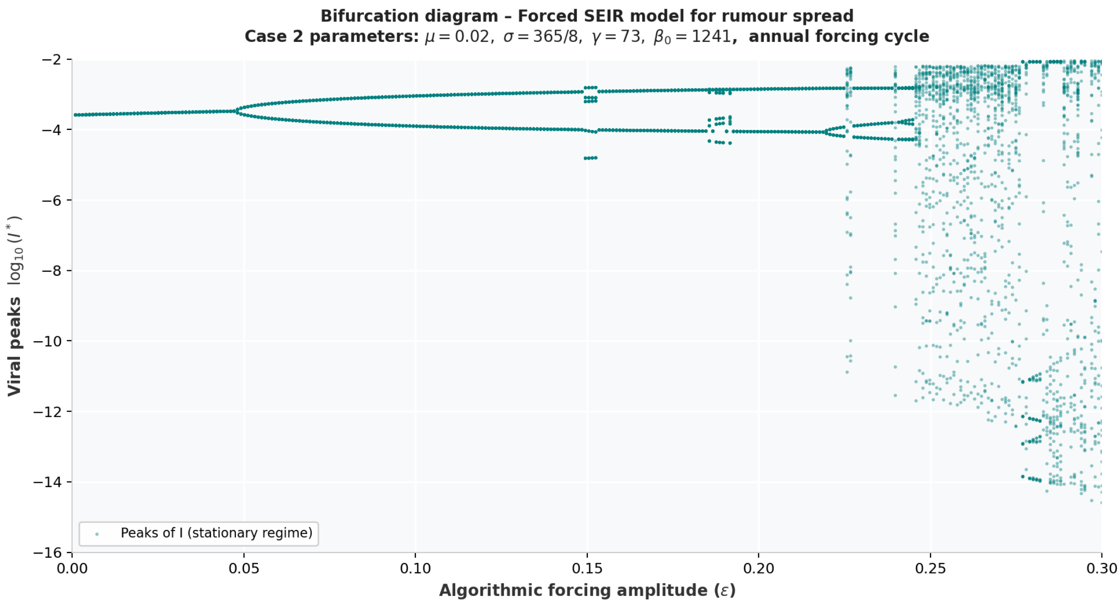

Fig. 4.2. Bifurcation diagram for Case 2 ($\gamma = 73, \sigma = 365/8, \mu = 0.02, \beta_0 = 1241$).

At low values of ε (where the algorithmic force is low), the system is stable, producing a single outbreak peak per year. However, at $\varepsilon \approx 0.05$, it begins to alternate between a strong and a weak peak. As the forcing amplitude grows, these branches split again near $\varepsilon \approx 0.15$, producing four distinct peak levels.

Above $\varepsilon \approx 0.25$, the periodic structure disappears, showing a dense cloud of points spanning over thirteen orders of magnitude (down to $\log_{10}(I^*) \approx -15$), suggesting the chaotic behaviour of the system.

This behaviour occurs because the engagement timescale (approximately five days) aligns the natural oscillation period of the system with the one year algorithmic forcing. This resonance allows each cycle to interact with the previous one, driving the instability that leads to chaos.

Case 3

Figure 4.3 shows the bifurcation diagram for Case 3.

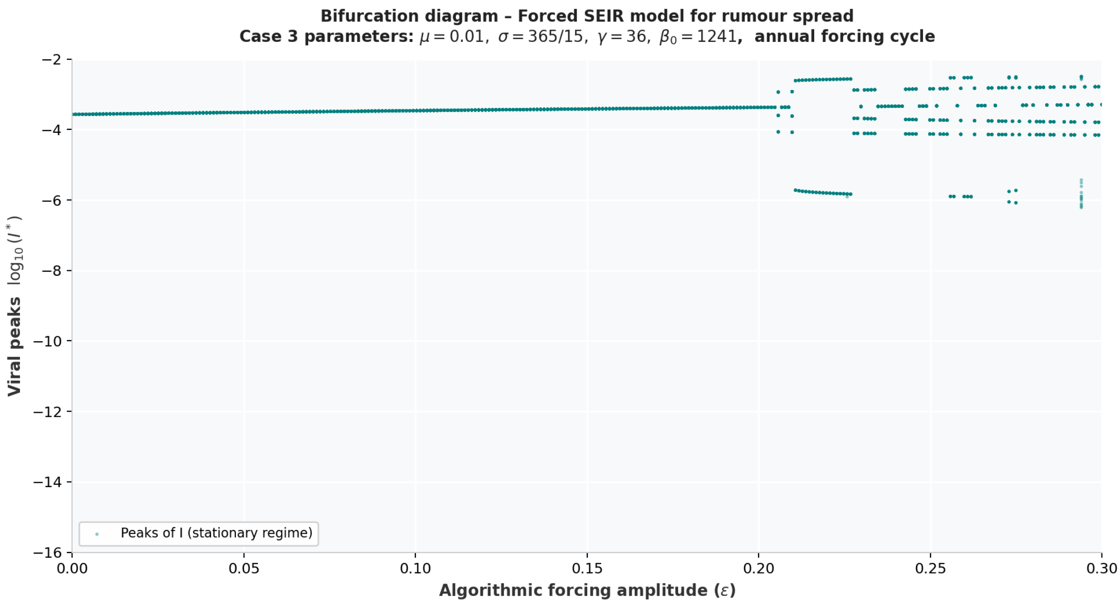

Fig. 4.3. Bifurcation diagram for Case 3 ($\gamma = 36$, $\sigma = 365/15$, $\mu = 0.01$, $\beta_0 = 1241$).

In this case, for low values of ε, the system remains stable and predictable until $\varepsilon \approx 0.20$.

Above $\varepsilon \approx 0.20$ two things happen simultaneously. The peak values spread across $\log_{10}(I^*) \in [-3, -2]$, indicating that outbreaks are now varying in intensity from cycle to cycle. At the same time, a separate cluster of points appears much lower in the diagram, around $\log_{10}(I^*) \approx -6$. The system is switching intermittently between two distinct regimes (normal outbreaks and almost extinction events) without any intermediate states.

This behaviour reflects the slower engagement timescale of approximately ten days. The natural oscillation period of this system is longer than one year, so the forcing resonance is weaker than in Case 2. The forcing is still strong enough at high ε to occasionally push the spreading population near zero.

Case 4

Figure 4.4 shows the bifurcation diagram for Case 4.

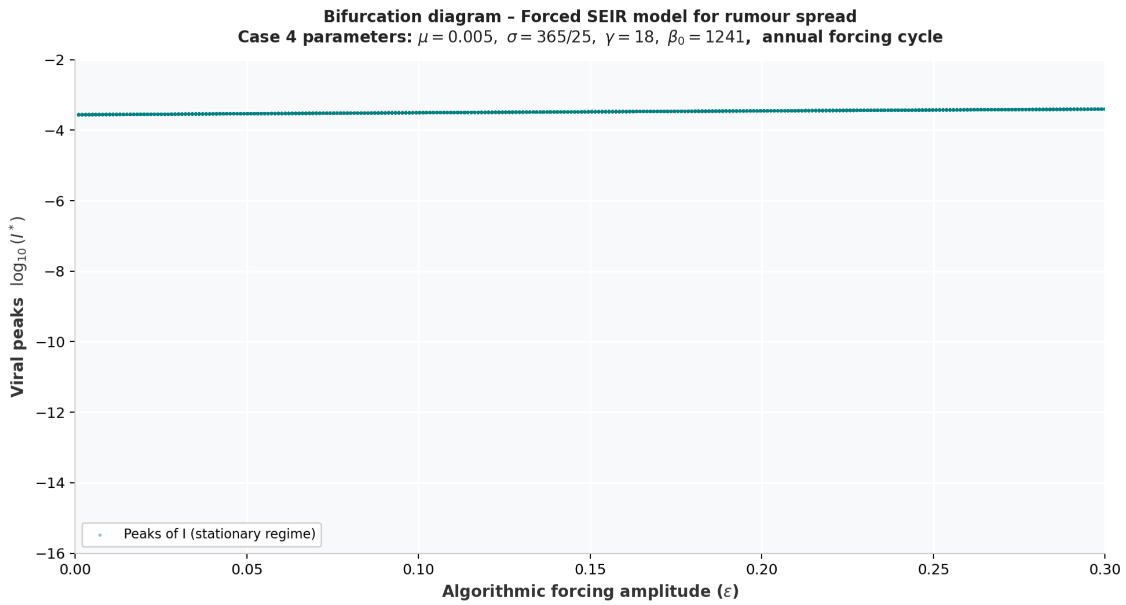

Fig. 4.4. Bifurcation diagram for Case 4 ($\gamma = 18, \sigma = 365/25, \mu = 0.005, \beta_0 = 1241$).

The bifurcation diagram for Case 4 is, in contrast to the other three, completely stable. Increasing ε from zero to its maximum value changes almost nothing about the dynamics.

This result shows that the same algorithm that destabilises Case 2 completely has no effect here. The reason is that users in this configuration remain active spreaders for approximately twenty days before disengaging. This makes the natural cycle of the spreading process much longer than one year, so the annual forcing period is too short to interact with the system.

4.2.4. Comparative discussion

Figure 4.5 places all four diagrams on a common axis in order to compare them.

First, the engagement timescale $1/\gamma$ is the parameter that most determines the structure of the bifurcation diagram. As γ decreases from 120 to 18, the oscillation period of the unforced system lengthens, and the resonance with the one-year forcing degrades.

Second, at high γ (Case 1) the chaos is intermittent and near-extinction, with occasional very small peaks corresponding to cycles where the rumour almost vanishes. At intermediate γ (Case 2) the chaos is more evident, spanning more than thirteen orders of magnitude in $\log_{10}(I^*)$. At low γ (Case 3) the chaos is intermittent in a different sense, producing a periodic cycle with a separate extinction cluster. Case 4 does not enter any irregular regime within the studied range.

Third, and most importantly for the interpretation of the model, the chaotic behaviour is not determined only by the level of spreading activity but by the resonance between the engagement dynamics and the forcing period. The same algorithm produces qualitatively

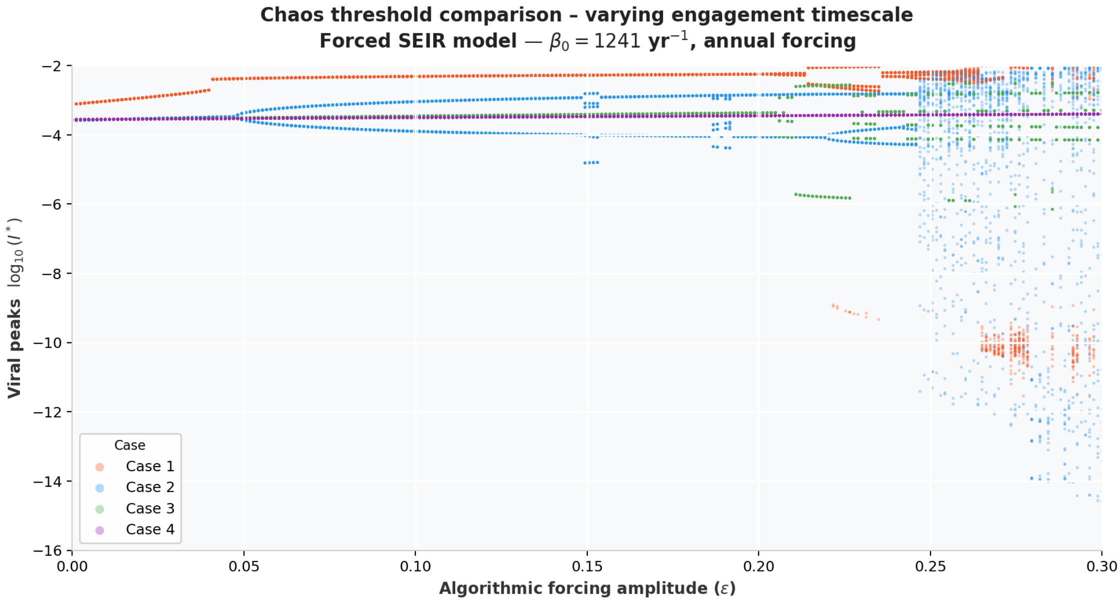

Fig. 4.5. Overlay of the four bifurcation diagrams on a common axis.

different outcomes depending only on user behaviour.

4.3. Maximal Lyapunov Exponent analysis

The bifurcation diagrams of the previous section provide a useful picture of the dynamical transitions in each case, but they cannot distinguish with certainty between true chaos and periodic cycles. Analysing the *Maximal Lyapunov Exponent* (MLE), we can clearly distinguish the chaotic behaviour in each case. As stated in Chapter 3, a strictly positive MLE is both necessary and sufficient for sensitive dependence on initial conditions, which is the defining property of chaos.

4.3.1. Methodology

The MLE is computed numerically for each of the four cases using the Benettin renormalisation algorithm described in Section 3.2 (see [2]). For each value of ε in the range $[0, 0.30]$ (120 equally spaced values), the system is integrated over a transient of 950 years to reach the stationary regime, after which the MLE is estimated over a window of 50 years using a renormalisation interval of $\tau = 0.01$ yr (≈ 3.65 days) and an initial perturbation of magnitude $\delta_0 = 10^{-8}$ applied to the susceptible compartment. Moreover, any renormalisation step in which the distance between the reference and shadow trajectories grows by more than a factor of 10^6 relative to δ_0 is discarded. This prevents occasional numerical errors from distorting the final average.

4.3.2. Results and analysis

The results are shown in Figures 4.6 through 4.9. The red points indicate values of ε for which the MLE is strictly positive, showing chaotic dynamics, and blue points indicate non-positive values, corresponding to stable regimes.

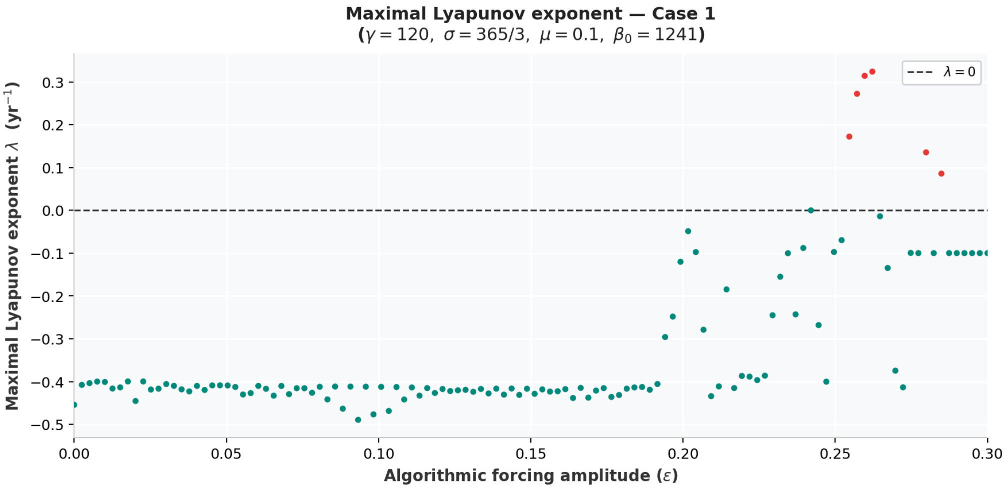

Fig. 4.6. Maximal Lyapunov exponent for Case 1.

The MLE remains negative for $\varepsilon \lesssim 0.20$, showing the stability of the period-doubling structure visible in the bifurcation diagram. Above $\varepsilon \approx 0.20$ the MLE fluctuates between -0.49 and $-0.05\,\mathrm{yr}^{-1}$, which reflects the irregular behaviour the system. Strictly positive values are observed at only 6 out of 120 sampled points, all concentrated in the interval $\varepsilon \in [0.255, 0.295]$, with a maximum of $\lambda_1 \approx 0.32\,\mathrm{yr}^{-1}$. Therefore, chaotic behaviour appears but is discontinuous, showing isolated windows of chaos close to ordered regime. This is consistent with the bifurcation diagram, where periodic an chaotic regimes become almost indistinguishable at high ε.

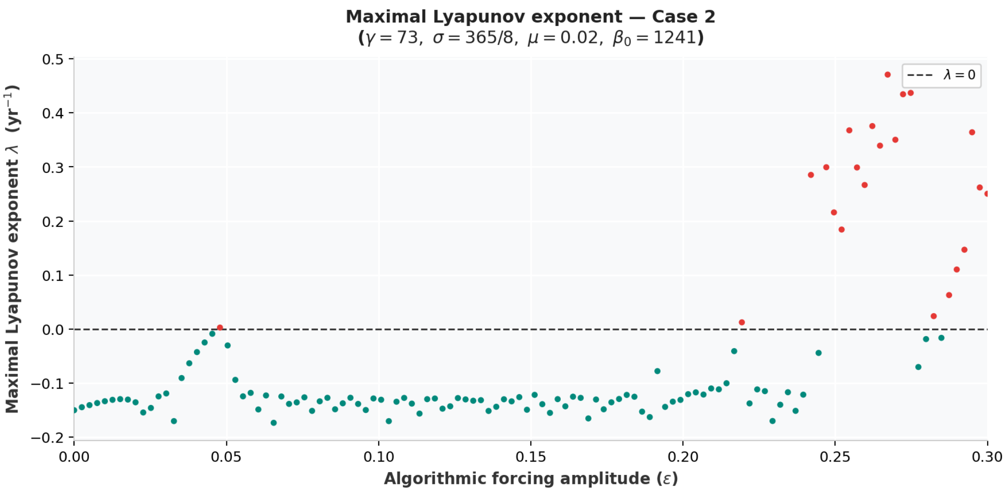

Fig. 4.7. Maximal Lyapunov exponent for Case 2.

In this case, the MLE rises monotonically from $-0.15\,\mathrm{yr}^{-1}$ at $\varepsilon = 0$ toward zero, crossing the chaos threshold for the first time at $\varepsilon \approx 0.048$. This coincides with the first period-doubling bifurcation identified in Figure 4.2, showing the instability of the transition from a period-1 to a period-2 orbit. The MLE then returns to negative values through the periodic regime, becomes positive again from $\varepsilon \approx 0.245$ onward, where it rises to a maximum of $\lambda_1 \approx 0.47\,\mathrm{yr}^{-1}$. In total, 22 out of 120 sampled values yield a positive MLE, all for high-ε values . So that chaotic behaviour is more extended in this case, consistent with the bifurcation diagram.

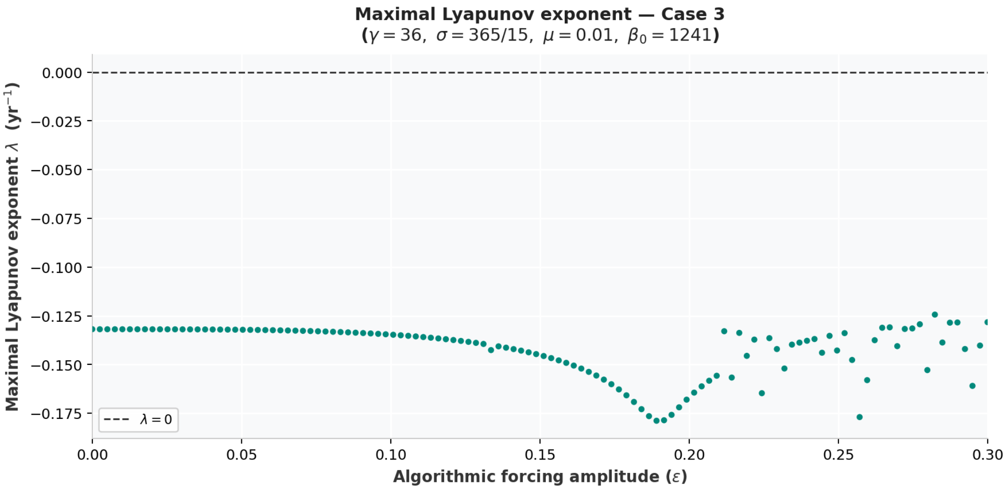

Fig. 4.8. Maximal Lyapunov exponent for Case 3.

The MLE is strictly negative for all 120 sampled values of ε. This clearly shows the stability among all the regimes observed in the bifurcation diagram. However, the MLE decreases gradually from $-0.130\,\mathrm{yr}^{-1}$ at $\varepsilon = 0$ to a minimum of $-0.178\,\mathrm{yr}^{-1}$ near $\varepsilon \approx 0.19$, and then increases again toward $-0.125\,\mathrm{yr}^{-1}$ for larger ε. The minimum of the MLE at $\varepsilon \approx 0.19$ coincides with the intermittent structure seen in the bifurcation diagram. This suggests that the behaviour observed in the bifurcation diagram above $\varepsilon \approx 0.20$ is a complex periodic attractor.

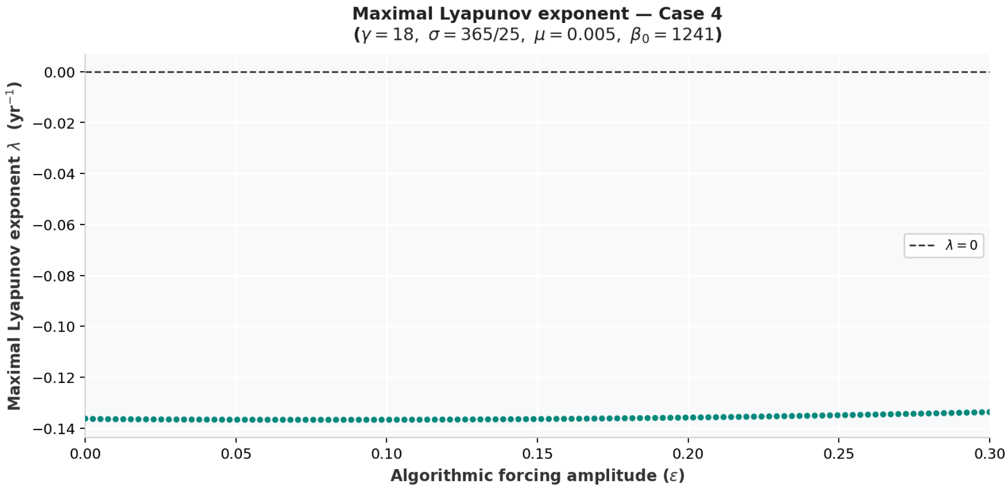

Fig. 4.9. Maximal Lyapunov exponent for Case 4.

Finally, here the MLE is constant at $\lambda_1 \approx -0.137\,\text{yr}^{-1}$ across the entire range, with negligible variation between the minimum and maximum sampled values. This shows that the dynamics of Case 4 are not affected at all by the algorithmic forcing, as the system remains on the same stable attractor regardless of ε.

4.3.3. Comparative discussion

Now we can characterise more precisely the dynamics of the 4 cases.

Table 4.2 summarises the key results.

Case	λ_1 at $\varepsilon = 0$	Chaos onset (ε)	Max λ_1	Chaotic points
Case 1	−0.42	0.255	+0.32	6 / 120
Case 2	−0.15	0.048	+0.47	22 / 120
Case 3	−0.13	n/a	−0.13	0 / 120
Case 4	−0.14	n/a	−0.14	0 / 120

TABLE 4.2. SUMMARY OF MAXIMAL LYAPUNOV EXPONENT RESULTS FOR THE FOUR CASES.

To sum up, after analysing both the bifurcation diagrams and the MLE, these results show clearly where chaotic regimes appear.

We can observe how Case 3 and Case 4 never enter into a chaotic regime. Moreover, although chaos in Case 2 was clear by looking at the bifurcation diagram, now we can also observe the exact values of ε where the MLE becomes positive, showing the chaotic windows.

Finally, Case 1 behaviour was not very clear in the bifurcation diagram, since the periodic and chaotic regimes become almost indistinguishable at high ε. However, the MLE analysis shows that there are indeed some isolated windows of chaos in this case, although they are not very extended.

As a final conclusion, the MLE analysis shows that the algorithmic forcing can produce chaotic dynamics in the forced SEIR system, but this effect is highly dependent on the engagement parameters of the system, which is consistent with the bifurcation diagram.

4.4. Phase portrait analysis

This section analyses in detail the geometry of the asymptotic dynamics of the forced SEIR system through phase portraits and time series. The analysis is restricted to Case 2 ($\gamma = 73$, $\sigma = 365/8$, $\mu = 0.02$, $\beta_0 = 1241$), which produces the most interesting dynamical behaviour across the range of forcing amplitudes studied.

4.4.1. Methodology

For each selected value of ε, the system is integrated numerically using `scipy.integrate.odeint` with absolute and relative tolerances of 10^{-17} and 10^{-8} respectively. The absolute tolerance is set intentionally strict to avoid numerical noise accumulation when computing population fractions over long periods. The integration spans a total time of 1000 years; the first 950 years are discarded as a transient, and the analysis is performed over the remaining 50 years, ensuring that the solution has converged to its asymptotic attractor.

Two trajectories are computed for each value of ε, differing only in the initial condition of the exposed compartment: $E(0) = 0.001$ and $E(0) = 0.01$, while keeping $S(0) = 0.06$ and $I(0) = 0.001$ fixed. The remaining parameters are those of Case 2, listed in Table 4.1. The comparison between the two trajectories shows the sensitivity to initial conditions in each case. If the two trajectories converge to the same attractor, the system is insensitive to this perturbation. On the other hand, if they diverge and explore different regions of phase space, the system could exhibit sensitive dependence on initial conditions, showing chaotic behaviour, or bistability, showing multiple coexisting attractors.

Each figure presents two panels. The left panel shows the three-dimensional phase portrait in the (S, E, I) space, with all variables plotted on a $\log_{10}$ scale to make the full dynamical range visible. The right panel shows the corresponding time series of the infected fraction $I(t)$ over the 50-year sampling window.

Four values of ε are examined, chosen to be representative of the distinct dynamical regimes identified in the bifurcation diagram of Case 2 (Figure 4.2): $\varepsilon = 0.10$, within the period-doubling regime; $\varepsilon = 0.19$, in the higher-period regime approaching the chaotic threshold; $\varepsilon = 0.25$, at the onset of chaos; and $\varepsilon = 0.28$, entering a periodic window.

Remark. A numerical analysis have been performed in order to check the accurate tolerance in the integrating method. Moreover, further analysis have been done in order to determine the differences between bistability and chaos in the phase portraits, since both of them can produce divergence between trajectories. The results of these analyses are not included in this chapter for simplicity, but they show the robustness of the results and can be checked in the Github repository found in the Appendix A.

4.4.2. Results and analysis

Case $\varepsilon = 0.10$

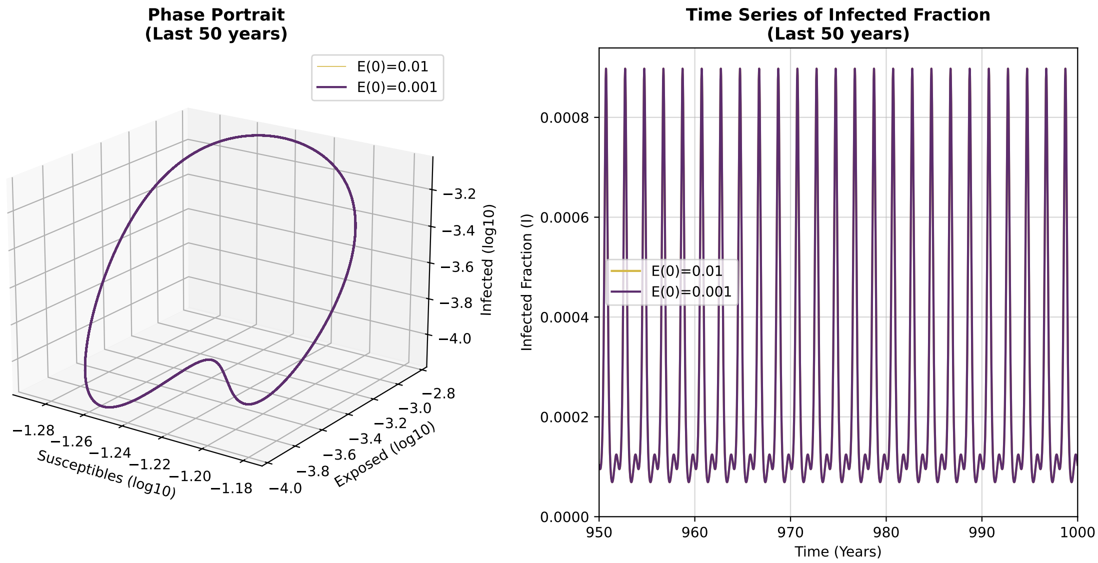

Fig. 4.10. Phase portrait and time series for Case 2 with $\varepsilon = 0.10$

At $\varepsilon = 0.10$, the system lies within the period-doubling regime identified in the bifurcation diagram, between the first bifurcation at $\varepsilon \approx 0.05$ and the second at $\varepsilon \approx 0.15$. The phase portrait shows two trajectories that converge to the same closed curve in the (S, E, I) space regardless of the initial condition, showing that the system converges to a stable periodic orbit.

The time series on the right panel reflects this structure clearly, as both trajectories settle into the same regular pattern of alternating strong and weak outbreaks, with a period of two years. This is consistent with the bifurcation diagram at this value of ε, and to the MLE value of $\lambda \approx -0.13$. As a conclusion, for this specific case, the system is in a stable periodic regime where the dynamics are predictable.

Case $\varepsilon = 0.19$

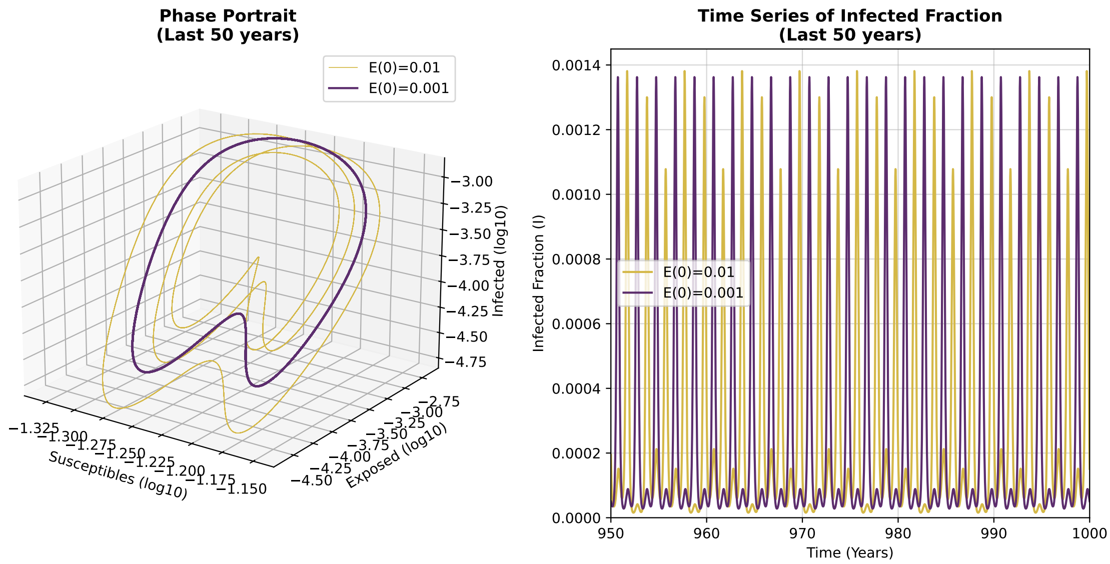

Fig. 4.11. Phase portrait and time series for Case 2 with $\varepsilon = 0.19$

At $\varepsilon = 0.19$, the system has undergone further period-doubling bifurcations and now enters a higher-period regime. The phase portrait shows that the asymptotic dynamics occupy a larger region of the (S, E, I) space than at $\varepsilon = 0.10$. Moreover, the trajectories trace out a different structure for each initial condition. Rather than converging to the exact same attractor, the system exhibits bistability, meaning that multiple stable periodic orbits coexist, and the final state depends on the initial number of exposed individuals. The attractors are no longer simple closed curves but have developed a more complex geometry.

The time series shows a repeating pattern with a longer period and more varied peak heights than at $\varepsilon = 0.10$, with each trajectory settling into a distinct sequence of outbreaks. Although the final path depends on the initial conditions, the system is not chaotic; it remains strictly deterministic and predictable, as the negative maximal Lyapunov exponent ($\lambda \approx -0.16$) confirms, but with a significantly more complex temporal structure.

Case $\varepsilon = 0.25$

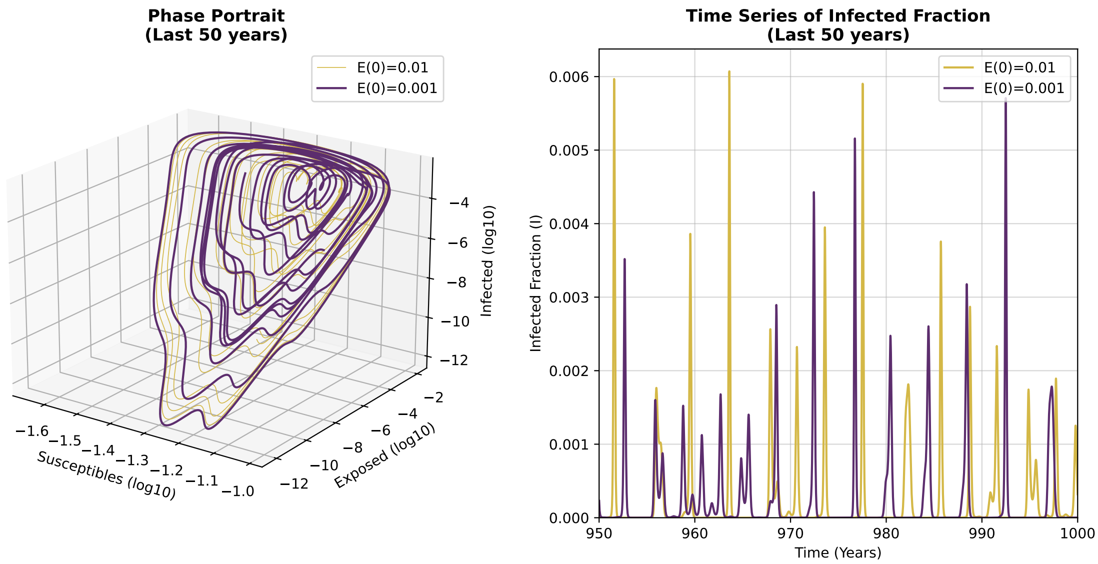

Fig. 4.12. Phase portrait and time series for Case 2 with $\varepsilon = 0.25$.

At $\varepsilon = 0.25$, the system has entered into the chaotic regime. The phase portrait shows a strange attractor, a complex and non-repeating trajectory that fills a bounded region of the (S, E, I) space without ever closing on itself. The two trajectories explore different regions of the attractor over the 50-year sampling window, showing that the system now exhibits sensitive dependence on initial conditions.

The time series on the right panel reinforces this interpretation. Both trajectories produce irregular sequences of outbreaks with highly variable peak heights, and the two sequences are visibly different from each other despite starting from nearby initial conditions. The system still produces recurrent outbreaks driven by the annual forcing cycle, but their intensity is no longer predictable from one year to the next. These results are consistent with the MLE analysis, which yielded a result of $\lambda \approx +0.21$ at this value of ε.

Case ε = 0.28

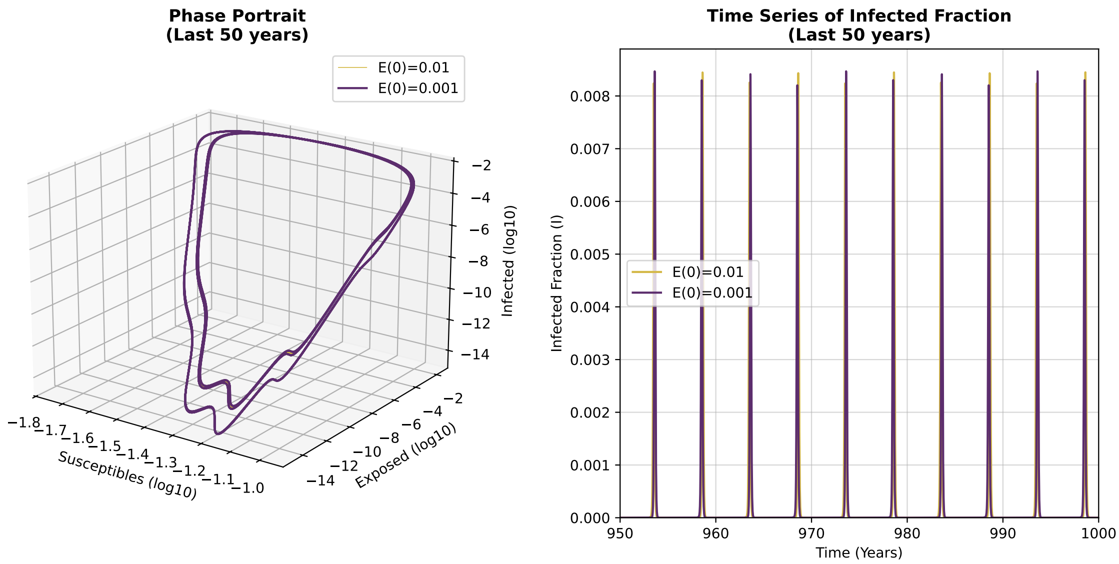

Fig. 4.13. Phase portrait and time series for Case 2 with $\varepsilon = 0.28$.

Finally, in this case the system enters a periodic window, where the MLE is negative and the dynamics are predictable.

We can observe how both trajectories converge to the same attractor in the phase space, showing that the system is not sensitive to initial conditions. The time series also shows similar periodic sequences of outbreaks.

This means that, although the system entered a chaotic region, the dynamics change to periodic for this specific value of ε. This is consistent with the MLE analysis, which showed a value of $\lambda \approx -0.0185$ at $\varepsilon \approx 0.2798$.

4.4.3. Comparative discussion

All together, this gives us a visualisation of how a rumour changes as an algorithm changes its force. It shows that, for this specific case, increasing algorithmic amplification completely reshapes how information flows through the user base.

At low amplification ($\varepsilon = 0.10$), the system settles into predictable loop. In the real world, this represents a manageable situation. A rumour might have its seasonal highs and lows, but it follows a regular, repeating pattern. Even as the forcing increases ($\varepsilon = 0.19$), and the system undergoes period-doubling, remains predictable.

The situation changes completely when the algorithmic forcing reaches $\varepsilon = 0.25$. At this point, the system becomes chaotic, and the predictable loops are replaced by a strange attractor. For a social media platform, this means the algorithm has amplified the content so aggressively that it breaks the rumour's natural spreading cycle. The main consequence is the system's sensitivity to initial conditions. In practice, if just a few extra users interact with the rumour today, the outcome months later could be entirely different (it might trigger a viral wave, or it might just die out). Because of this unpredictability, the platform completely loses the ability to control the rumour's impact.

However, not for all high values of ε the system is chaotic. For example, at $\varepsilon = 0.28$ the system enters a periodic window, where the MLE is negative and the dynamics are predictable. This shows that, even if the system enters a chaotic regime, it can jump to stable cycles with small changes in the forcing amplitude.

5. REGULATORY FRAMEWORK AND SOCIOECONOMIC IMPACT

This chapter addresses the regulatory and socioeconomic context of the work presented in this thesis.

5.1. Regulatory framework

Although this work is a theoretical and computational study in mathematics, several regulatory considerations are relevant.

This thesis is an original academic work. All mathematical results are either derived from first principles or properly attributed to their sources via bibliographic references, in compliance with Spanish intellectual property law (Real Decreto Legislativo 1/1996, de 12 de abril, por el que se aprueba el texto refundido de la Ley de Propiedad Intelectual).

All computational scripts are original implementations written in Python, an open-source programming language distributed under the PSF Licence. The libraries used — NumPy, SciPy, and Matplotlib — are distributed under open-source licences (BSD and PSF), and their use in this work respects their respective licence terms.

Furthermore, Chapter 4 proposes a mathematical model for the spread of rumours on social media platforms. Although the model is purely theoretical and does not involve the collection or processing of real user data, it is directly relevant to the European regulatory landscape on platform accountability. The Digital Services Act (Regulation (EU) 2022/2065), in force since 2024, establishes obligations for online platforms regarding algorithmic transparency and the mitigation of systemic risks associated with the amplification of harmful content. The results of Chapter 4 provide a theoretical framework that supports such regulation.

5.2. Socioeconomic environment and sustainable development goals

This section analyses the socioeconomic environment of this thesis and its relation to the United Nations Sustainable Development Goals (SDGs).

From a scientific perspective, the mathematical tools developed in this thesis have a wide range of practical applications across disciplines. In meteorology, chaos theory forms the basis of modern ensemble weather forecasting systems, which have a direct economic impact by improving the accuracy of predictions used in agriculture, energy, transport, and disaster management. In medicine, the study of chaotic dynamics in cardiac and neural systems has opened new diagnostic and therapeutic avenues. In finance, the recognition of non-linear behaviour in markets has influenced risk modelling and the design of more robust financial instruments. In engineering, chaos-based encryption protocols are used in secure communications. These applications generate significant economic value and have a positive impact on society.

The specific application developed in Chapter 4, a forced SEIR model for the spread of rumours on social media platforms, has direct socioeconomic implications. The model shows that algorithmic amplification can drive information dynamics into chaotic regimes, making the impact of content promotion strategies unpredictable. This result has practical implications for the design of content moderation policies and platform regulation, and supports the idea behind the Digital Services Act (Regulation (EU) 2022/2065). However, it is important to note that the model is purely theoretical and has not been validated against real data, which limits the scope of its conclusions.

Regarding the environmental impact of this work, it is worth noting that all simulations were carried out on a standard personal laptop, with a negligible carbon footprint. No data centres or cloud computing resources were used. The environmental impact of this thesis is therefore considered minimal.

With respect to the United Nations Sustainable Development Goals, this work is most directly related to the following:

SDG 4: Quality Education. This thesis contributes to the advancement of mathematical knowledge and its dissemination at university level. By presenting a rigorous treatment of chaos theory and non-linear dynamical systems, it serves as an educational resource for students and researchers interested in these topics.

SDG 9: Industry, Innovation and Infrastructure. The mathematical modelling techniques developed in this work are directly applicable to a wide range of engineering and industrial problems, including the design of control systems, secure communications, and predictive models for complex systems. The application to social media dynamics also contributes to the development of transparent algorithmic infrastructure.

SDG 16: Peace, Justice and Strong Institutions. The model proposed in Chapter 4 directly addresses the problem of misinformation on digital platforms, which is increasingly recognised as a threat to democratic institutions and social cohesion. By providing a mathematical framework for understanding how algorithmic amplification destabilises information dynamics, this work contributes to the broader effort to design fairer and more transparent online platforms.

SDG 17: Partnerships for the Goals. This thesis draws on an interdisciplinary approach, combining tools from pure mathematics, computational science, and epidemiological modelling to address a problem of social relevance. This kind of cross-disciplinary collaboration is essential for addressing complex global challenges.

6. PROJECT PLANNING AND BUDGET

This chapter presents the planning and budget of this thesis.

6.1. Project planning

The project was carried out over approximately ten months, from September 2025 to June 2026, and can be divided into four phases.

The first phase, running from September to December 2025, was dedicated to reading and documentation. The primary references were studied in depth during this period, together with a review of the supporting mathematical literature. This phase provided the theoretical foundation for the rest of the work.

The second phase, from January to March 2026, was dedicated to the theoretical development of the thesis. This covered the writing of Chapter 1, on the introduction and motivation of the work, Chapter 2, on the basic concepts of non-linear dynamical systems, and Chapter 3, on chaos theory, including the analysis of the Lorenz system, the logistic map, Lyapunov exponents, fractal geometry, and strange attractors. All Python scripts for the figures in these chapters were also developed during this period.

The third phase, from April to May 2026, was dedicated to the computational model analysed in Chapter 4. This included the design and implementation of the forced SEIR model for rumour spread on social media, the bifurcation analysis, the Lyapunov exponent computation, and the phase portrait analysis, together with the writing of the corresponding sections.

The fourth and final phase, during the first two weeks of June 2026, was dedicated to the consolidation of the project. This included the writing of Chapters 5 and 6, the revision and correction of the full document following tutor feedback.

Figure 6.1 illustrates the project timeline.

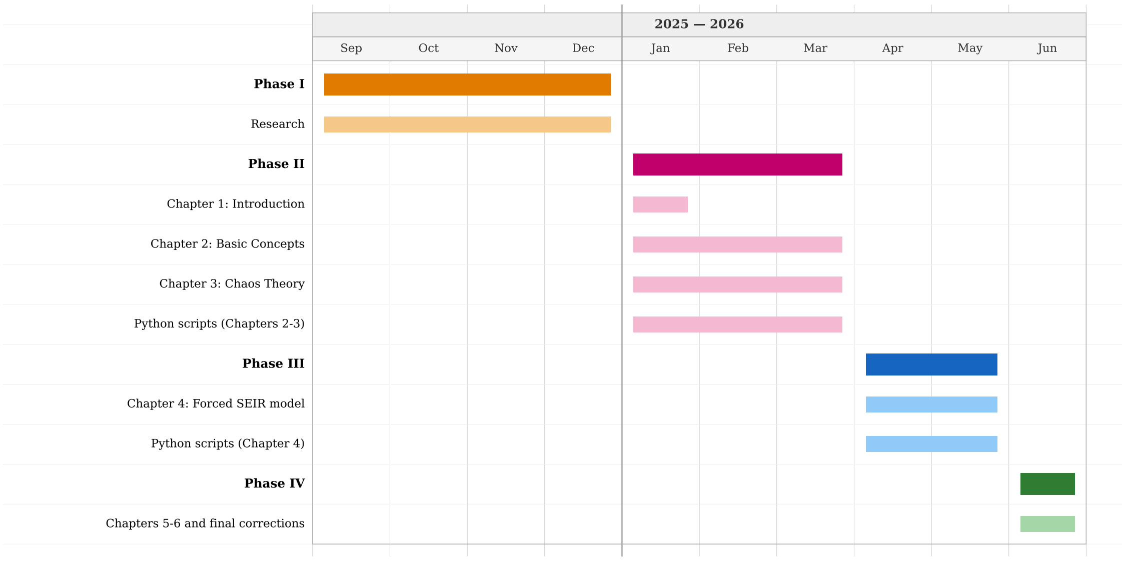

Fig. 6.1. Project timeline.

6.2. Financial allocation

Regarding human resources, two profiles are considered. The student is treated as a junior researcher. According to the III Convenio Colectivo del Personal Laboral de Investigación, the minimum gross annual salary for this category is 25,892 € [16]. After applying the corresponding income tax rate (IRPF) of 19.42% and a social security contribution of 8.75%, the net annual salary is approximately 18,593 €. Assuming a standard 40-hour workweek and 12 months without bonuses, the estimated net hourly rate is approximately $18{,}593/(12 \times 4 \times 40) \approx 9.69$ €/hour. The thesis supervisor is a full professor, with a gross annual salary of approximately 47,500 € [19], corresponding to a gross hourly rate of approximately 24.74 €/hour.

According to the UC3M guidelines, the Bachelor's Thesis is equivalent to 12 ECTS, with each ECTS credit corresponding to 25 hours of student work, giving a total of 300 hours. At the estimated hourly rate of 9.69 €, the total cost of student work is $300 \times 9.69 = 2{,}907$ €. The supervisor participated in approximately 10 tutorial sessions of one hour each, plus an estimated 15 additional hours dedicated to document revisions and corrections, totalling 25 hours of supervision at a cost of $25 \times 24.74 = 618.50$ €.

Regarding material resources, the references used in this work were accessed through public libraries at no additional cost. All computational work was carried out in Python, an open-source programming language available at no cost. The thesis was written in LaTeX using Overleaf, whose free plan was sufficient for the development of this work. A Lenovo IdeaPad laptop was used for all simulations and document preparation, with an estimated market price of 650 € [12]. Claude Pro, an AI assistant used for code debugging, was used for 6 months at a cost of approximately 22.40 €/month (including 21% VAT applied in Spain [11]), for a total of 134.40 €.

The following table summarises the estimated costs associated with this project.

Resource	**Hours**	**Cost (€)**
Student (junior researcher, 9.69 €/h)	300	2,907.00
Supervisor (full professor, 24.74 €/h)	25	618.50
Lenovo IdeaPad laptop	—	650.00
Claude Pro (6 months, 22.40 €/month)	—	134.40
Primary references (UC3M library)	—	0
Python + NumPy + SciPy + Matplotlib	—	0
Overleaf (free plan)	—	0
Total	**325**	**4,309.90**

TABLE 6.1. ESTIMATED BUDGET FOR THE DEVELOPMENT OF THIS THESIS.

7. CONCLUSIONS

7.1. Summary and social interpretation

This thesis has explored the wonderful world of chaos theory and non-linear dynamical systems, with a particular focus on their applications to real world problems. Altogether, this thesis has highlighted the importance of the study non-linear dynamical systems in different scientific disciplines, and how the mathematical tools developed in this thesis can be applied to understand complex behaviours present in nature and society.

Regarding the proposed model, all the analyses perfomed show how both algorithms in social platforms together with user behaviour can lead to a wide variety of dynamical behaviours. This has a direct impact in the way information is spread, and can be dangerous for certain types of content, such as misinformation or harmful rumours.

The bifurcation diagrams and the MLE analysis show the importance of user behaviour, as the same algorithm leads to completely different outcomes on the dynamics of rumour spreading. Therefore, this shows the importance of understanding user behaviour and engagement patterns when designing algorithms for content promotion.

Finally, the phase portrait analysis for Case 2, where widely chaotic regimes appear, show how dangerous can be algorithm amplification for specific types of users or platforms.

Altogether, it is clear that, although the model lacks empirical validation, it provides a useful theoretical framework to understand the complex dynamics of information spreading under algorithmic influence, and highlights the potential risks of aggressive amplification strategies combined with different user profiles.

7.2. Limitations of the model

While the proposed forced SEIR model provides useful insights into how recommendation algorithms destabilise rumour spreading, it is important to outline its limitations.

- **Lack of empirical validation with real data.** This model is exploratory, based on past analogous such as the SIR model used for rumour spreading. Although the parameter values (such as the engagement rates γ and σ) were chosen to represent realistic social media behaviours, the model has not been calibrated against real world data. Obtaining accurate data from platforms is challenging due to privacy restrictions and the proprietary nature of their algorithms. Therefore, while the mathematical conclusions regarding chaos and resonance are valid for the parameters analysed, the exact numerical results remain theoretical and would need to be validated against actual data.

- **Simplification of the algorithmic forcing function.** To study the system using the classical tools of dynamical systems theory, the algorithmic amplification was modelled as a continuous function, $\beta(t) = \beta_0(1 - \varepsilon\cos(2\pi t))$. Approximating this complex behaviour with a single annual periodic function is a necessary mathematical simplification, but it does not behave exactly as real digital algorithms.

- **Lack of sociological and network validation.** The SEIR model treats the platform's user base as a homogeneous population. It assumes that every user has the same probability of encountering the rumour and reacts to it in the same way. In reality, the sociological dynamics of social media are more complex. The model does not take into account isolated communities, or polarisation. Furthermore, it ignores the impact of users with a notably larger activity (like influencers) and individual psychology like confirmation bias.

Nevertheless, despite these limitations, the model provides an insight that the classical SIR model does not, including algorithm amplification as a main factor in rumour spreading.

7.3. Future research

There are many ways of extending and improving the model developed.

As mentioned in the limitations, the model analysed in this work treats the user population as a single homogeneous group. In reality, social media users are organised into communities, interest groups or geographical regions that interact with each other but have distinct engagement characteristics. An extension of the model would be to couple several SEIR compartments, each representing a different community with its own parameters γ_i, σ_i and μ_i, and connected to the others through inter-community transmission terms that model the flow of content across different communities.

Moreover, it may be of interest to analyse the spread of rumours across multiple platforms simultaneously. In the present model, users belong to a single platform with a single algorithmic forcing cycle. In practice, the same rumours often circulates across several platforms at once, each with its own recommendation algorithm, its own user base, and its own engagement timescale. A user exposed to a rumour on one platform may become an active spreader on another, creating a cross-platform transmission that this model cannot represent.

Finally, the model could be analysed for different (and more realistic) parameter values if there existed the possibility of calibrating it with real data, as well as trying different approaches such as iterated maps that could model the system efficiently.

BIBLIOGRAPHY

[1] Sitaram Asur, Bernardo A Huberman, Gabor Szabo, and Chunyan Wang. "Trends in social media: Persistence and decay". In: *Proceedings of the International AAAI Conference on Web and Social Media*. Vol. 5. 1. 2011, pp. 434–437.

[2] G. Benettin, L. Galgani, A. Giorgilli, and J.-M. Strelcyn. "Lyapunov Characteristic Exponents for Smooth Dynamical Systems and for Hamiltonian Systems; a Method for Computing All of Them. Part 1: Theory". In: *Meccanica* 15.1 (1980), pp. 9–20.

[3] Luís MA Bettencourt, Ariel Cintrón-Arias, David I Kaiser, and Carlos Castillo-Chávez. "The power of a good idea: Quantitative modeling of the spread of ideas from epidemiological models". In: *Physica A: Statistical Mechanics and its Applications* 364 (2006), pp. 513–536.

[4] Tania Cambero Jiménez and Roberto Barrio Gil. "Modelos matemáticos de difusión de enfermedades: influencia de perturbaciones". In: *Universidad de Zaragoza* (2019).

[5] Robert Devaney. *An introduction to chaotic dynamical systems*. CRC press, 2018.

[6] Kenneth Falconer. *Fractals: A very short introduction*. OUP Oxford, 2013.

[7] Mitchell J Feigenbaum. "Quantitative universality for a class of nonlinear transformations". In: *Journal of statistical physics* 19.1 (1978), pp. 25–52.

[8] James Gleick. *Chaos: Making a new science*. Penguin, 2008.

[9] Peter Grassberger and Itamar Procaccia. "Characterization of strange attractors". In: *Physical review letters* 50.5 (1983), p. 346.

[10] Ferenc Huszár, Sofia Ira Ktena, Conor O'Brien, Luca Belli, Andrew Schlaikjer, and Moritz Hardt. "Algorithmic amplification of politics on Twitter". In: *Proceedings of the national academy of sciences* 119.1 (2022), e2025334119.

[11] Javadex. *Claude AI en Español: Planes, Precios y Límites*. Accessed: 06/06/2026. 2026. URL: https://www.javadex.es/blog/claude-ai-gratis-espanol-planes-precios-limites-guia-2026.

[12] Lenovo. *Comprar portátiles IdeaPad | Lenovo España*. Accessed: 06/06/2026. 2025. URL: https://www.lenovo.com/es/es/d/ideapad-ultra-thin.

[13] Tien-Yien Li and James A Yorke. "Period three implies chaos". In: *The american mathematical monthly* 82.10 (1975), pp. 985–992.

[14] Edward N Lorenz. "Deterministic nonperiodic flow 1". In: *Universality in Chaos, 2nd edition*. Routledge, 2017, pp. 367–378.

[15] Benoit B Mandelbrot. "The fractal geometry of nature/Revised and enlarged edition". In: *New York* (1983).

[16] Ministerio de Trabajo y Economía Social. *III Convenio Colectivo del Personal Laboral de Investigación*. Resolución de 16 de mayo de 2024, BOE-A-2024-10663. 2024.

[17] Meksianis Z Ndii, Ema Carnia, and Asep K Supriatna. "Mathematical models for the spread of rumors: a review". In: *Issues and trends in interdisciplinary behavior and social science* (2018), pp. 65–73.

[18] Edward Ott. *Chaos in dynamical systems*. Cambridge university press, 2002.

[19] Remitly. *¿Cuánto gana un profesor universitario en España?* Accessed: 06/06/2026. 2025. URL: https://www.remitly.com/blog/es/educacion/sueldo-profesor-universidad/.

[20] Colin Sparrow. *The Lorenz equations: bifurcations, chaos, and strange attractors*. Springer Science & Business Media, 2012.

[21] Steven H Strogatz. *Nonlinear dynamics and chaos: with applications to physics, biology, chemistry, and engineering*. Chapman and Hall/CRC, 2024.

[22] K Thompson, R Castro Estrada, D Daugherty, A Cintron-Arias, et al. "A deterministic approach to the spread of rumors". In: (2003).

[23] Fang Wu and Bernardo A Huberman. "Novelty and collective attention". In: *Proceedings of the National Academy of Sciences* 104.45 (2007), pp. 17599–17601.

APPENDIX A: CODE REPOSITORY

All Python scripts used to generate the figures in this thesis are publicly available in a GitHub repository. The repository is organised into three folders, one per chapter, each containing the corresponding scripts and a `results/` subfolder where the output figures are saved.

The repository can be accessed at the following URL:

https://github.com/laaumartin/Chaos_Theory.git

It includes a file named `README.md` containing a description of the contents and instructions for running the various simulation codes implemented along the project.

APPENDIX B: GENERATIVE AI USAGE DECLARATION

The following pages include the official declaration form required by the university.

DECLARATION OF USE OF GENERATIVE ARTIFICIAL INTELLIGENCE (AI) IN BACHELOR THESIS (TFG)

I have used Generative AI in my TFG
Check all that apply:

YES

If you have ticked YES, please complete the following 3 parts of this document:

Part 1: Reflection on legal, ethical and responsible behavior

Please be aware that the use of AI carries some risks and may lead to a series of serious academic consequences: the evaluation of your TFG could be compromised if the use of AI involves the use of confidential data, copyrighted materials, or personal data, used without complying with the conditions required in each case (authorization of the interested parties, authorization of the owners, following the instructions of the University).

Question
1. In my interactions with AI tools, I have provided **confidential data**, always with the appropriate authorization of the data subjects. Confidentiality encompasses any information that a person or organization wishes to protect for legal, commercial, privacy, or strategic reasons (such as patents or trade secrets).
NO, I have not used confidential data
2. In my interaction with Generative AI tools, I have submitted **copyrighted materials** with the permission of those concerned.
NO, I have not used copyrighted materials
3. In my interaction with Generative AI tools, I have submitted **personal data** with the consent of the data subjects.

NO, I have not used personal data
4. My use of the Generative AI tool has **respected its terms of use**, as well as the essential ethical principles, not being maliciously oriented to obtain an inappropriate result for the work presented, that is to say, one that produces an impression or knowledge contrary to the reality of the results obtained, that supplants my own work or that could harm people.
YES

Part 2: Declaration of technical use

Use the following model statement as many times as necessary, in order to reflect all types of iteration you have had with Generative AI tools. Include one example for each type of use where indicated: *[Add an example].*

I declare that I have made use of the Generative AI system Claude Pro for:

Documentation and drafting:

- *Supporting in analysing alternatives and approaches using AI*

I have used the assistant as support to look for different interesting parameter configurations when analysing the developed model, as well as to detect errors in the modeling process.

- *Revision or rewriting of previously drafted paragraphs*

I have used the assistant as support for correcting grammatical errors, since English is not my primary language.

Develop specific content

Generative AI has been used as a support tool for the development of the specific content of the TFG, including:

- *Assistance in the development of lines of code (programming)*

I have used the assistant as a support tool for writing Python scripts (using libraries such as NumPy, SciPy, and Matplotlib) necessary to generate the bifurcation diagrams, calculate the

Lyapunov exponents and to generate plots. Additionally, I used it for writing LaTeX code, since I did not know how to write some complex mathematical expressions correctly.

- *Optimisation processes*

I used the tool for code debugging. For example, I asked for help identifying and correcting logical and syntax errors in the Python scripts when doing the numerical simulations of the dynamical systems.

Part 3: Reflection on utility

Please provide a personal assessment (free format) of the strengths and weaknesses you have identified in the use of Generative AI tools in the development of your work. Mention if it has helped you in the learning process, or in the development or drawing conclusions from your work.

Thanks to the AI assistant, I have been able to develop a much more interesting project, since it has helped me for the most time demanding parts, such as code debugging, correcting grammatical errors or finding interesting parameter configurations to analyse.

However, the generated content has been validated previously both from the author of this thesis and the supervisor, so no mathematical or technical mistakes were made. Moreover, all the examples, models and techniques have been exclusively proposed from books and articles, not by this assistant. In this way, the assistant has only helped by putting into practice in the code these techniques previously studied by professionals.